

# COMPLETE DATA ANALYTICS COURSE

*From Beginner to Job-Ready Data Analyst*

Data Analytics • Excel • SQL • Statistics • Python • Pandas • NumPy • Data Visualization • Power BI • Tableau • Data Cleaning • Exploratory Data Analysis • Real-World Projects

A complete, self-study course book covering the full data analytics toolkit — theory, formulas, real datasets, practical exercises, projects, and interview preparation.

**DATA-ANALYTICS-HUB**

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## CHAPTER 1

# Introduction to Data Analytics

### Learning Objectives

- Understand what data, data analytics and data analysis mean.
- Distinguish Data Analytics from Data Science and Business Intelligence.
- Learn the four types of analytics and the data analytics lifecycle.
- Understand the role, skills and career path of a Data Analyst.

## 1.1 What is Data?

Data is any raw fact, figure, or observation collected about people, events, transactions, or processes. On its own, data has no meaning until it is organized and interpreted. For example, the numbers 25, 30, 42 mean nothing alone, but if we say 'the ages of three customers are 25, 30 and 42', the numbers become meaningful information.

Data can come from many sources: sales transactions, website clicks, sensors, surveys, social media, and business applications. Every action in a modern business — a purchase, a login, a support call — generates data.

## 1.2 What is Data Analytics?

Data Analytics is the science of examining raw data to draw conclusions, identify patterns, and support decision-making. It combines statistics, computer science, and business knowledge to convert data into actionable insight.

### Real-World Example

An online store notices that sales fall every Tuesday. A data analyst studies transaction data, discovers Tuesday delivery delays caused by a logistics partner, and recommends a schedule change — data analytics turned a number into a business decision.

## 1.3 What is Data Analysis?

Data Analysis is the specific process of inspecting, cleaning, and modeling data to discover useful information. Data Analytics is the broader discipline (the field, tools, and strategy); Data Analysis is one activity within it — the hands-on examination of a dataset.

## 1.4 Data Analytics vs Data Science

| Aspect | Data Analytics                       | Data Science                               |
|--------|--------------------------------------|--------------------------------------------|
| Focus  | Analyzing existing data for insights | Building models to predict future outcomes |
| Tools  | Excel, SQL, Power BI, Tableau        | Python, R, Machine Learning frameworks     |

| Aspect           | Data Analytics                 | Data Science                              |
|------------------|--------------------------------|-------------------------------------------|
| Skill depth      | Statistics, business reporting | Statistics, ML, programming, deep math    |
| Output           | Dashboards, reports, insights  | Predictive models, algorithms, AI systems |
| Typical question | 'What happened and why?'       | 'What will happen next?'                  |

## 1.5 Data Analytics vs Business Intelligence

Business Intelligence (BI) focuses on monitoring what is happening in a business right now through dashboards and standard reports, usually using historical and current data. Data Analytics goes further — it explores deeper questions, tests hypotheses, and can involve statistical and predictive techniques beyond a fixed dashboard.

## 1.6 Why Data Analytics is Important

- Helps organizations make evidence-based decisions instead of guesses.
- Identifies trends, risks, and opportunities early.
- Improves operational efficiency by highlighting bottlenecks.
- Enhances customer experience through personalization.
- Provides a competitive advantage in nearly every industry.

## 1.7 Types of Data Analytics

| Type                   | Question Answered           | Example                             |
|------------------------|-----------------------------|-------------------------------------|
| Descriptive Analytics  | What happened?              | Monthly sales report                |
| Diagnostic Analytics   | Why did it happen?          | Sales dropped because of a stockout |
| Predictive Analytics   | What is likely to happen?   | Forecasting next quarter's demand   |
| Prescriptive Analytics | What should we do about it? | Recommending reorder quantities     |

These four types build on one another: descriptive analytics summarizes the past, diagnostic analytics explains it, predictive analytics forecasts the future, and prescriptive analytics recommends the best action to take.

## 1.8 Data Analytics Lifecycle

- 1 **Business Understanding** — define the problem and objective.
- 2 **Data Collection** — gather data from relevant sources.
- 3 **Data Cleaning** — remove errors, duplicates, and missing values.
- 4 **Data Exploration** — study patterns and relationships (EDA).
- 5 **Data Analysis** — apply statistical or analytical techniques.
- 6 **Data Visualization** — present findings using charts and dashboards.
- 7 **Interpretation & Communication** — explain insights to stakeholders.
- 8 **Decision Making** — use insights to take business action.

## 1.9 Role of a Data Analyst

A Data Analyst collects, cleans, and studies data to answer specific business questions. They build reports and dashboards, run statistical checks, and communicate findings to managers, marketing teams, or product teams so that decisions are backed by evidence rather than opinion.

## 1.10 Skills Required for a Data Analyst

| Category    | Skills                                             |
|-------------|----------------------------------------------------|
| Technical   | Excel, SQL, Python/R basics, Power BI or Tableau   |
| Analytical  | Statistics, critical thinking, problem structuring |
| Business    | Domain knowledge, KPI understanding, storytelling  |
| Soft skills | Communication, presentation, attention to detail   |

## 1.11 Career Opportunities

- Business Analyst, Data Analyst, Reporting Analyst
- BI Developer / BI Analyst
- Marketing / Sales / Financial Analyst
- Operations Analyst, Product Analyst
- Junior Data Scientist (as a growth path)

## 1.12 Real-World Applications

| Industry            | Use of Data Analytics                            |
|---------------------|--------------------------------------------------|
| Retail / E-commerce | Demand forecasting, personalized recommendations |
| Banking / Finance   | Fraud detection, credit risk scoring             |
| Healthcare          | Patient outcome analysis, resource planning      |
| Marketing           | Campaign performance, customer segmentation      |
| Logistics           | Route optimization, delivery time prediction     |

### Key Takeaways

- Data becomes information only after it is processed and interpreted.
- Data Analytics is a broad discipline; Data Analysis is the hands-on task within it.
- There are four types of analytics: descriptive, diagnostic, predictive, prescriptive.
- The Data Analytics Lifecycle moves from business understanding to decision-making.
- A Data Analyst needs a mix of technical, analytical, and communication skills.

## Chapter Summary

This chapter introduced the foundation of data analytics: what data is, how analytics differs from data science and BI, the four types of analytics, the end-to-end lifecycle, and the skills and career paths open to a data analyst. These concepts form the base on which every later chapter — Excel, SQL, Python, statistics, and visualization — is built.

## Key Terms

- **Data** — raw, unprocessed facts and figures.
- **Information** — data that has been processed to be meaningful.
- **Insight** — an actionable conclusion drawn from information.
- **KPI** — Key Performance Indicator, a measurable business metric.
- **Dashboard** — a visual summary of key metrics and trends.

## Practice Questions

- 1 Define data, information, and insight with one example each.
- 2 Explain the difference between Data Analytics and Data Science in your own words.
- 3 List the four types of analytics and give a business example of each.
- 4 Describe the eight stages of the Data Analytics Lifecycle.
- 5 Name three industries that rely heavily on data analytics and explain how.

## Interview Questions

### Q1. What is the difference between Data Analytics and Business Intelligence?

BI focuses on monitoring current business performance through standard dashboards and reports, while Data Analytics involves deeper exploration, hypothesis testing, and sometimes predictive techniques to answer open questions, not just track known metrics.

### Q2. What are the four types of data analytics?

Descriptive (what happened), diagnostic (why it happened), predictive (what will happen), and prescriptive (what should be done).

### Q3. What does a Data Analyst do on a typical day?

Collects and cleans data, writes SQL queries or Excel formulas, builds reports/dashboards, runs basic statistical checks, and presents findings to stakeholders to support decisions.

## CHAPTER 2

# Types of Data and Data Sources

### Learning Objectives

- Differentiate structured, semi-structured, and unstructured data.
- Understand qualitative vs quantitative and discrete vs continuous data.
- Distinguish primary/secondary and internal/external data.
- Identify common real-world data sources used by analysts.

## 2.1 Structured, Semi-Structured and Unstructured Data

| Type            | Description                                       | Example                           |
|-----------------|---------------------------------------------------|-----------------------------------|
| Structured      | Organized in rows and columns with a fixed schema | SQL database table, Excel sheet   |
| Semi-Structured | Has some organization but no strict schema        | JSON, XML, email headers          |
| Unstructured    | No predefined format or organization              | Images, videos, free-text reviews |

Roughly 80% of business data is unstructured or semi-structured. Analysts typically convert it into a structured form (tables) before analysis.

## 2.2 Qualitative vs Quantitative Data

| Aspect   | Qualitative Data                 | Quantitative Data                 |
|----------|----------------------------------|-----------------------------------|
| Nature   | Descriptive, categorical         | Numeric, measurable               |
| Example  | Customer feedback, color, gender | Age, sales amount, temperature    |
| Analysis | Themes, frequencies, categories  | Averages, sums, statistical tests |

## 2.3 Discrete vs Continuous Data

**Discrete data** can only take specific, countable values — for example, the number of orders placed (1, 2, 3...).

**Continuous data** can take any value within a range, including decimals — for example, height, weight, or time taken to deliver a package (2.35 hours).

## 2.4 Categorical vs Numerical Data

- **Categorical data** — represents groups or labels, e.g. 'Region: North, South, East, West'.
- **Numerical data** — represents measurable quantities, e.g. 'Revenue: 45,230'.
- Categorical data is further split into **nominal** (no order, e.g. gender) and **ordinal** (has order, e.g. rating: Low, Medium, High).

## 2.5 Primary vs Secondary Data

| Type           | Definition                                     | Example                                 |
|----------------|------------------------------------------------|-----------------------------------------|
| Primary Data   | Collected first-hand for a specific purpose    | A survey you design and run yourself    |
| Secondary Data | Collected by someone else, reused for analysis | Government census data, public datasets |

## 2.6 Internal vs External Data

**Internal data** originates within the organization — sales records, HR data, CRM entries. **External data** comes from outside sources — market research reports, social media, weather data, or competitor pricing. Analysts often combine both to get a complete picture.

## 2.7 Common Data Sources

| Source                  | Typical Use                                                |
|-------------------------|------------------------------------------------------------|
| Relational Databases    | Transactional business data (orders, customers)            |
| Excel / CSV Files       | Manually maintained or exported reports                    |
| APIs                    | Live data feeds from external services (weather, currency) |
| Websites (Web Scraping) | Public data not available via API                          |
| Surveys                 | First-hand feedback and opinion data                       |
| Business Applications   | CRM, ERP, POS systems                                      |
| Cloud Platforms         | Data lakes/warehouses such as AWS S3, BigQuery, Snowflake  |

### Key Takeaways

- Structured data fits neatly into tables; unstructured data does not.
- Qualitative data describes qualities; quantitative data measures quantities.
- Discrete data is countable; continuous data can take any value in a range.
- Primary data is collected first-hand; secondary data is reused from elsewhere.
- Real-world analysis often blends internal and external data sources.

## Practice Questions

- 1 Classify the following as structured, semi-structured, or unstructured: a CSV file, a customer support chat log, a JSON API response.
- 2 Give two examples each of discrete and continuous data from everyday life.
- 3 Explain why most business data starts unstructured and how it becomes usable.
- 4 Identify whether 'customer satisfaction rating (1-5)' is nominal or ordinal.

## Interview Questions

**Q1. What is the difference between structured and unstructured data?**

Structured data follows a fixed schema of rows and columns (like a database table), while unstructured data — such as images or free text — has no predefined organization and must be processed before analysis.

**Q2. Give an example of semi-structured data.**

JSON or XML files are semi-structured: they have tags or key-value pairs that add some organization, but not a strict tabular schema.

**Q3. What is the difference between primary and secondary data?**

Primary data is collected directly by the analyst for a specific purpose (e.g. a new survey); secondary data already exists and is reused, such as publicly published datasets.

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## CHAPTER 3

# The Data Analytics Process

### Learning Objectives

- Learn the complete ten-step data analytics workflow.
- Apply the workflow to a realistic e-commerce business example.
- Understand how each stage feeds into the next.

## 3.1 The End-to-End Workflow

Every analytics project — whether it takes one day or three months — follows roughly the same sequence of stages. Skipping a stage (especially data cleaning) is one of the most common reasons analyses produce wrong conclusions.

- 1 **Define the business problem** — what decision needs to be supported?
- 2 **Collect data** — identify and gather relevant data sources.
- 3 **Understand data** — review structure, size, and data types.
- 4 **Clean data** — fix missing values, duplicates, and errors.
- 5 **Transform data** — reshape, merge, and standardize for analysis.
- 6 **Explore data** — look for patterns, distributions, and outliers.
- 7 **Analyze data** — apply statistical or business logic to answer the question.
- 8 **Visualize data** — turn findings into charts and dashboards.
- 9 **Communicate insights** — present clear, actionable findings.
- 10 **Make business decisions** — the organization acts on the insight.

## 3.2 Worked Example: An E-Commerce Company

Consider 'ShopEase', an online retailer whose management notices that profit has fallen over the last quarter despite stable sales. Here is how the ten-step process applies:

| Stage              | Applied to ShopEase                                                |
|--------------------|--------------------------------------------------------------------|
| 1. Define problem  | Why did profit fall while revenue stayed flat?                     |
| 2. Collect data    | Pull order, discount, and shipping data from the database          |
| 3. Understand data | Check columns: Order ID, Sales, Discount, Profit, Region           |
| 4. Clean data      | Remove duplicate orders, fix negative-quantity errors              |
| 5. Transform data  | Merge orders with product cost data to compute margins             |
| 6. Explore data    | Plot profit by region and category to spot weak spots              |
| 7. Analyze data    | Find that discounts above 30% turn several categories unprofitable |

| Stage             | Applied to ShopEase                            |
|-------------------|------------------------------------------------|
| 8. Visualize data | Build a bar chart of profit by discount band   |
| 9. Communicate    | Present a one-page summary to the pricing team |
| 10. Decide        | Cap discounts at 25% on low-margin categories  |

### Key Insight

Notice that the technical steps (cleaning, transforming, analyzing) are only useful because they are anchored to a clear business question defined in step 1. Analysts who jump straight to charts without a defined problem often produce interesting but unusable output.

### Key Takeaways

- The analytics process always starts with a clear business question.
- Data cleaning and transformation typically consume the most analyst time.
- Visualization and communication are what turn analysis into decisions.
- The same ten-step process scales from a one-page report to a full project.

## Practice Questions

- 1 Pick a business problem (e.g. rising customer complaints) and map it to the ten-step analytics process.
- 2 Why is 'define the business problem' placed before data collection?
- 3 What could go wrong if the data cleaning step is skipped?

## Interview Questions

### Q1. Walk me through your typical analytics workflow.

I start by clarifying the business question with stakeholders, then collect and understand the data, clean and transform it, explore it for patterns, run the analysis, visualize the results, and present clear, actionable recommendations.

### Q2. Which step of the analytics process usually takes the most time?

Data cleaning and transformation — real-world data is rarely ready for analysis and often needs significant preparation before it can be trusted.

## CHAPTER 4

# Excel for Data Analytics

### Learning Objectives

- Navigate the Excel interface and understand data types.
- Clean and organize data using sorting, filtering, and validation.
- Master essential formulas: lookup, logical, text, and aggregate functions.
- Build Pivot Tables and a simple dashboard.

## 4.1 The Excel Interface

Excel organizes data into a **workbook** made of **worksheets**. Each worksheet is a grid of **rows** (numbered) and **columns** (lettered); the intersection of a row and column is a **cell**, identified by a reference such as B3. Every analysis in Excel starts by understanding what data type each column holds — text, number, date, or boolean.

## 4.2 Core Data Preparation Tools

| Tool                   | Purpose                                                      |
|------------------------|--------------------------------------------------------------|
| Sort                   | Arrange rows by one or more columns, ascending or descending |
| Filter                 | Show only rows matching selected criteria                    |
| Remove Duplicates      | Delete repeated rows based on chosen columns                 |
| Find & Replace         | Locate and correct values across a sheet                     |
| Text to Columns        | Split one column into several using a delimiter              |
| Data Validation        | Restrict cell input to a defined list or rule                |
| Conditional Formatting | Highlight cells automatically based on rules                 |
| Excel Tables           | Convert a range into a structured, auto-expanding table      |

## 4.3 Essential Excel Formulas

For every important formula: Syntax, a plain explanation, a worked example, the expected result, and a common mistake to avoid.

### SUM

**Syntax:** =SUM(range)

**Explanation:** Adds all numbers in a range.

**Example:** =SUM(A2:A10)    **Result:** Total of A2:A10

**Common mistake:** Including text cells accidentally returns 0 for them, not an error.

## AVERAGE

**Syntax:** =AVERAGE(range)

**Explanation:** Calculates the mean of a range.

**Example:** =AVERAGE(B2:B10)    **Result:** Mean of B2:B10

**Common mistake:** Blank cells are ignored, but 0s are included and can skew the result.

## COUNT

**Syntax:** =COUNT(range)

**Explanation:** Counts numeric cells only.

**Example:** =COUNT(C2:C10)    **Result:** Count of numbers

**Common mistake:** Does not count text — use COUNTA for that.

## COUNTA

**Syntax:** =COUNTA(range)

**Explanation:** Counts non-empty cells (any type).

**Example:** =COUNTA(C2:C10)    **Result:** Count of filled cells

**Common mistake:** Counts cells with a stray space as 'filled'.

## MIN / MAX

**Syntax:** =MIN(range) / =MAX(range)

**Explanation:** Returns the smallest / largest value.

**Example:** =MAX(D2:D10)    **Result:** Largest value in D2:D10

**Common mistake:** Ignores text; won't flag non-numeric data entry errors.

## IF

**Syntax:** =IF(condition, value\_if\_true, value\_if\_false)

**Explanation:** Returns one value or another based on a test.

**Example:** =IF(E2>100,"High","Low")    **Result:** "High" or "Low"

**Common mistake:** Forgetting quotes around text results causes a #NAME? error.

## AND / OR

**Syntax:** =AND(c1,c2) / =OR(c1,c2)

**Explanation:** Combines multiple logical conditions.

**Example:** =IF(AND(A2>0,B2>0),"OK","Check")    **Result:** "OK" or "Check"

**Common mistake:** Mixing up AND/OR reverses your intended logic.

## IFERROR

**Syntax:** =IFERROR(formula, value\_if\_error)

**Explanation:** Replaces error results with a custom value.

**Example:** =IFERROR(A2/B2,0) **Result:** 0 instead of #DIV/0!

**Common mistake:** Hides real errors if used too broadly without checking cause.

## SUMIF

**Syntax:** =SUMIF(range, criteria, sum\_range)

**Explanation:** Sums values that meet one condition.

**Example:** =SUMIF(C2:C10,"North",D2:D10) **Result:** Total sales for 'North'

**Common mistake:** Criteria range and sum range must be the same size.

## SUMIFS

**Syntax:** =SUMIFS(sum\_range, crit\_range1, crit1, ...)

**Explanation:** Sums values meeting multiple conditions.

**Example:** =SUMIFS(D2:D10,C2:C10,"North",E2:E10,">100") **Result:** Filtered total

**Common mistake:** Sum range comes first here, unlike SUMIF.

## COUNTIF / COUNTIFS

**Syntax:** =COUNTIF(range, criteria)

**Explanation:** Counts cells meeting one or more conditions.

**Example:** =COUNTIFS(C2:C10,"North",E2:E10,">100") **Result:** Count matching both conditions

**Common mistake:** Text criteria need quotes; numeric comparisons need operators in quotes too.

## AVERAGEIF

**Syntax:** =AVERAGEIF(range, criteria, avg\_range)

**Explanation:** Averages values meeting a condition.

**Example:** =AVERAGEIF(C2:C10,"North",D2:D10) **Result:** Average sales for 'North'

**Common mistake:** Returns #DIV/0! if no rows match the criteria.

## VLOOKUP

**Syntax:** =VLOOKUP(value, table, col\_index, FALSE)

**Explanation:** Looks up a value in the first column of a range.

**Example:** =VLOOKUP(A2,Products!A:C,3,FALSE) **Result:** Matched value from column 3

**Common mistake:** Always set the last argument to FALSE for an exact match.

## XLOOKUP

**Syntax:** =XLOOKUP(value, lookup\_array, return\_array)

**Explanation:** Modern, flexible lookup in any direction.

**Example:** =XLOOKUP(A2,Products!A:A,Products!C:C) **Result:** Matched value

**Common mistake:** Not available in older Excel versions.

## INDEX + MATCH

**Syntax:** =INDEX(range, MATCH(value, lookup\_range, 0))

**Explanation:** Flexible two-part lookup, works left or right.

**Example:** =INDEX(C:C,MATCH(A2,B:B,0)) **Result:** Matched value from column C

**Common mistake:** MATCH's third argument must be 0 for exact match.

## LEFT / RIGHT / MID

**Syntax:** =LEFT(text,n) etc.

**Explanation:** Extracts characters from a text string.

**Example:** =LEFT(A2,3) **Result:** First 3 characters

**Common mistake:** Counts characters, not words.

## LEN

**Syntax:** =LEN(text)

**Explanation:** Returns the number of characters in text.

**Example:** =LEN(A2) **Result:** Character count

**Common mistake:** Counts leading/trailing spaces too.

## TRIM

**Syntax:** =TRIM(text)

**Explanation:** Removes extra spaces from text.

**Example:** =TRIM(A2) **Result:** Cleaned text

**Common mistake:** Does not remove single spaces between words, only extras.

## CONCAT / TEXT

**Syntax:** =CONCAT(a,b) / =TEXT(value,"format")

**Explanation:** Joins text, or formats a value as text.

**Example:** =CONCAT(A2," ",B2) **Result:** Combined text

**Common mistake:** TEXT() output can no longer be used in numeric calculations.

## 4.4 Date Functions

- **TODAY()** — returns the current date.
- **YEAR() / MONTH() / DAY()** — extract parts of a date.
- **DATEDIF(start,end,"d")** — calculates the difference between two dates.
- **EOMONTH(date, months)** — returns the last day of a given month.

## 4.5 Pivot Tables and Pivot Charts

A **Pivot Table** summarizes large datasets by dragging fields into Rows, Columns, Values, and Filters areas — for example, dragging 'Region' to Rows and 'Sales' to Values instantly produces total sales per region without writing a single formula. A **Pivot Chart** visualizes a Pivot Table and updates automatically when the underlying data changes.

## 4.6 Building an Excel Dashboard

- 1 Structure raw data as an Excel Table.
- 2 Build one or more Pivot Tables to summarize key metrics.
- 3 Add Pivot Charts for trend and comparison visuals.
- 4 Add slicers to let users filter by Region, Category, or Date.
- 5 Arrange charts and KPI cells on a single 'Dashboard' sheet.
- 6 Apply consistent formatting and remove gridlines for a clean look.

## Excel Mini-Project: Regional Sales Report

### Mini-Project

- Dataset: an orders table with Order ID, Date, Region, Category, Sales, Discount, and Profit columns.
- Task 1: Use SUMIFS to calculate total sales per Region and Category.
- Task 2: Use IF and IFERROR to flag orders with a profit margin below 10%.
- Task 3: Build a Pivot Table summarizing Profit by Region and Month.
- Task 4: Create a Pivot Chart showing the sales trend over time.
- Task 5: Add a slicer for Region and assemble a one-page dashboard.
- Expected Learning Outcome: confidently combine formulas, Pivot Tables, and charts into a working, interactive Excel report.

### Key Takeaways

- SUMIF/COUNTIF/AVERAGEIF handle one condition; the \*IFS versions handle several.
- XLOOKUP and INDEX+MATCH are more flexible than VLOOKUP for real-world lookups.
- Pivot Tables summarize large datasets instantly without manual formulas.
- A good dashboard combines Pivot Tables, charts, and slicers on one clean sheet.

## Practice Questions

- 1 Write a formula using SUMIFS to total 'Profit' where Region is 'East' and Category is 'Furniture'.
- 2 Explain the difference between VLOOKUP and XLOOKUP.
- 3 What is the purpose of Data Validation, and give one real use case.
- 4 Describe the steps to build a Pivot Table from an orders dataset.

## Interview Questions

### Q1. What is the difference between VLOOKUP and INDEX+MATCH?

VLOOKUP only searches left-to-right and breaks if columns are inserted, while INDEX+MATCH can look in any direction and is more resilient to structural changes in the sheet.

### Q2. When would you use SUMIFS instead of SUMIF?

SUMIFS is used when a sum needs to satisfy two or more conditions simultaneously, such as summing sales for a specific region AND category.

### Q3. What is a Pivot Table used for?

It quickly summarizes and aggregates large datasets — for example totals, averages, or counts grouped by category — without writing manual formulas.

### Q4. How do you remove duplicate records in Excel?

Select the data range, go to Data > Remove Duplicates, and choose which columns should be checked for duplicate values.

# Statistics for Data Analytics

## Learning Objectives

- Understand population, sample, and core measures of central tendency.
- Calculate variance, standard deviation, and quartiles step by step.
- Understand correlation, probability basics, and distributions.
- Grasp hypothesis testing: null/alternative hypotheses and p-values.

## 5.1 Population and Sample

A **population** is the entire group being studied (e.g. all customers of a company). A **sample** is a smaller subset drawn from the population, used because studying an entire population is often impractical. A **variable** is any characteristic being measured, such as age, income, or purchase amount.

## 5.2 Measures of Central Tendency

| Measure | Definition                           | Formula / Method                         |
|---------|--------------------------------------|------------------------------------------|
| Mean    | The arithmetic average               | Sum of values ÷ number of values         |
| Median  | The middle value when data is sorted | Middle value (or average of two middles) |
| Mode    | The most frequently occurring value  | Value with highest frequency             |

## Numerical Example

Dataset: 4, 8, 6, 5, 3, 8, 9. Sorted: 3, 4, 5, 6, 8, 8, 9. Mean =  $(4+8+6+5+3+8+9)/7 = 43/7 = 6.14$ . Median = 6 (middle value). Mode = 8 (appears twice).

## 5.3 Range, Variance, and Standard Deviation

### Formula

Range = Maximum – Minimum | Variance ( $\sigma^2$ ) =  $\Sigma(x - \text{mean})^2 \div n$  | Standard Deviation ( $\sigma$ ) =  $\sqrt{\text{Variance}}$

Standard deviation measures how spread out the data is around the mean. A small standard deviation means values cluster tightly around the mean; a large one means values are widely spread.

## Step-by-Step Calculation

Dataset: 2, 4, 4, 4, 5, 5, 7, 9 (mean = 5). Deviations squared: 9, 1, 1, 1, 0, 0, 4, 16 → sum = 32. Variance =  $32 \div 8 = 4$ . Standard deviation =  $\sqrt{4} = 2$ .

## 5.4 Percentiles, Quartiles, and IQR

**Percentiles** indicate the value below which a given percentage of data falls. **Quartiles** split sorted data into four equal parts: Q1 (25th percentile), Q2 (median, 50th percentile), and Q3 (75th percentile). The **Interquartile Range (IQR)** =  $Q3 - Q1$ , and is commonly used to detect outliers.

#### Formula

Outlier rule: a value is an outlier if it is below  $Q1 - 1.5 \times IQR$  or above  $Q3 + 1.5 \times IQR$ .

## 5.5 Distribution and the Normal Distribution

A **distribution** describes how values in a dataset are spread. The **Normal Distribution** (bell curve) is symmetric around the mean, where most values cluster near the center. Many statistical tests assume the underlying data is approximately normal.

## 5.6 Skewness and Outliers

- **Right (positive) skew** — a long tail on the right; mean > median.
- **Left (negative) skew** — a long tail on the left; mean < median.
- **Outliers** — extreme values far from the rest of the data, which can distort the mean and should be investigated rather than blindly removed.

## 5.7 Correlation and Covariance

**Correlation** measures the strength and direction of a linear relationship between two variables, ranging from  $-1$  (perfect negative) to  $+1$  (perfect positive), with  $0$  meaning no linear relationship. **Covariance** indicates the direction of the relationship but, unlike correlation, is not standardized, so its scale depends on the units of the variables.

#### Business Example

A correlation of  $+0.85$  between advertising spend and sales suggests that as spend increases, sales tend to increase strongly — but correlation alone does not prove that advertising caused the sales increase.

## 5.8 Probability Basics

#### Formula

$P(\text{Event}) = \text{Number of favorable outcomes} \div \text{Total number of outcomes}$

**Conditional probability** is the probability of an event occurring given that another event has already occurred, written  $P(A|B)$  — for example, the probability a customer buys a laptop bag given they already bought a laptop.

## 5.9 Sampling and Sampling Methods

| Method          | Description                                                     |
|-----------------|-----------------------------------------------------------------|
| Random Sampling | Every member of the population has an equal chance of selection |

| Method               | Description                                                      |
|----------------------|------------------------------------------------------------------|
| Stratified Sampling  | Population divided into groups; sampled proportionally from each |
| Systematic Sampling  | Every nth member is selected from a list                         |
| Convenience Sampling | Easiest-to-reach members are selected (higher bias risk)         |

## 5.10 Confidence Intervals

A **confidence interval** gives a range within which the true population value is likely to fall, at a stated confidence level (commonly 95%). For example, 'average customer spend is \$120 ± \$8 at 95% confidence' means we are 95% confident the true average lies between \$112 and \$128.

## 5.11 Hypothesis Testing

- **Null Hypothesis (H0)** — the default assumption of no effect or no difference.
- **Alternative Hypothesis (H1)** — the claim we want to test evidence for.
- **p-value** — the probability of observing the data (or more extreme) if H0 were true; a small p-value (commonly < 0.05) suggests rejecting H0.
- **Type I Error** — incorrectly rejecting a true null hypothesis (false positive).
- **Type II Error** — incorrectly failing to reject a false null hypothesis (false negative).
- **Statistical significance** — a result unlikely to have occurred by chance alone.

### Applied Example

A company tests a new checkout page (H1: it increases conversion) against the old page (H0: no difference). If the test returns  $p = 0.02$ , this is below the 0.05 threshold, so the company rejects H0 and concludes the new page likely does improve conversion.

### Key Takeaways

- Mean, median, and mode summarize the center of data; each behaves differently with outliers.
- Standard deviation quantifies how spread out data is around the mean.
- Correlation shows relationship strength, not causation.
- Hypothesis testing uses a p-value to decide whether to reject the null hypothesis.

## Practice Questions

- 1 Given the dataset 10, 12, 12, 15, 18, 20, calculate the mean, median, and mode.
- 2 Calculate the variance and standard deviation for the dataset 3, 7, 7, 19.
- 3 Explain, in your own words, why correlation does not imply causation.
- 4 A p-value of 0.03 is obtained from a test using a 0.05 threshold. Should the null hypothesis be rejected? Explain why.

## Interview Questions

**Q1. What is the difference between variance and standard deviation?**

Variance is the average of squared deviations from the mean, expressed in squared units; standard deviation is the square root of variance, expressed in the original units, which makes it easier to interpret.

**Q2. What does a p-value tell you?**

It tells you the probability of observing results as extreme as the ones found, assuming the null hypothesis is true. A small p-value suggests the observed effect is unlikely to be due to chance.

**Q3. What is the difference between Type I and Type II errors?**

A Type I error is rejecting a true null hypothesis (a false alarm); a Type II error is failing to reject a false null hypothesis (a missed effect).

**Q4. How would you detect outliers in a dataset?**

Common approaches include the IQR rule (values beyond  $1.5 \times \text{IQR}$  from  $Q1/Q3$ ), z-scores beyond  $\pm 3$ , or visual methods such as box plots.

DATA-ANALYTICS-HUB

## CHAPTER 6

# SQL for Data Analytics

### Learning Objectives

- Understand relational databases, tables, keys, and SQL syntax.
- Write SELECT, WHERE, GROUP BY, HAVING, and JOIN queries confidently.
- Use subqueries, CTEs, and window functions for advanced analysis.
- Apply SQL to realistic customers/orders/products tables.

## 6.1 What is SQL?

SQL (Structured Query Language) is the standard language for storing, retrieving, and manipulating data in a relational **database**, which is managed by a **DBMS** (Database Management System) such as MySQL, PostgreSQL, or SQL Server. A database is organized into **tables**, each made of **rows** (records) and **columns** (fields).

- **Primary Key** — a column that uniquely identifies each row in a table.
- **Foreign Key** — a column that references the primary key of another table, creating a relationship between tables.

## 6.2 Sample Tables Used in This Chapter

| Table     | Key Columns                                            |
|-----------|--------------------------------------------------------|
| customers | customer_id (PK), name, region, signup_date            |
| orders    | order_id (PK), customer_id (FK), order_date, amount    |
| products  | product_id (PK), product_name, category, price         |
| employees | employee_id (PK), name, department, salary, manager_id |

### Q1 — Basic SELECT

```
SELECT name, region FROM customers;
```

**Explanation:** Retrieves the name and region columns for every customer.

**Expected Output:** A list of all customers with their region.

**Business Interpretation:** Gives a quick view of where customers are located.

### Q2 — Filtering with WHERE

```
SELECT * FROM orders WHERE amount > 500;
```

**Explanation:** Returns all columns for orders where the amount exceeds 500.

**Expected Output:** All high-value orders.

**Business Interpretation:** Helps identify large transactions for follow-up or review.

### Q3 — Sorting with ORDER BY

```
SELECT * FROM orders ORDER BY order_date DESC;
```

**Explanation:** Sorts orders from most recent to oldest.

**Expected Output:** Orders listed newest first.

**Business Interpretation:** Useful for reviewing the latest activity.

### Q4 — Aggregation with GROUP BY

```
SELECT region, SUM(amount) AS total_sales
FROM orders o JOIN customers c ON o.customer_id = c.customer_id
GROUP BY region;
```

**Explanation:** Joins orders to customers, then sums sales for each region.

**Expected Output:** One row per region with total sales.

**Business Interpretation:** Shows which regions generate the most revenue.

### Q5 — Filtering Groups with HAVING

```
SELECT customer_id, SUM(amount) AS total_spent
FROM orders
GROUP BY customer_id
HAVING SUM(amount) > 1000;
```

**Explanation:** Groups orders by customer, then keeps only customers spending over 1000.

**Expected Output:** High-value customers and their total spend.

**Business Interpretation:** Identifies customers worth prioritizing for loyalty programs.

### Q6 — DISTINCT and LIMIT

```
SELECT DISTINCT region FROM customers LIMIT 5;
```

**Explanation:** Returns up to 5 unique region values.

**Expected Output:** A short list of distinct regions.

**Business Interpretation:** Useful for quickly checking what categories exist in a column.

### Q7 — Aggregate Functions

```
SELECT COUNT(*) AS total_orders, AVG(amount) AS avg_order,  
MIN(amount) AS min_order, MAX(amount) AS max_order  
FROM orders;
```

**Explanation:** Calculates count, average, minimum, and maximum order amount in one query.

**Expected Output:** A single summary row with four statistics.

**Business Interpretation:** Gives a quick statistical snapshot of order sizes.

## Q8 — Pattern Matching with LIKE

```
SELECT * FROM customers WHERE name LIKE 'A%';
```

**Explanation:** Finds customers whose name starts with 'A'.

**Expected Output:** All customers with names beginning with A.

**Business Interpretation:** Useful for text search and data audits.

## Q9 — IN and BETWEEN

```
SELECT * FROM orders  
WHERE region IN ('North', 'East')  
AND amount BETWEEN 100 AND 500;
```

**Explanation:** Filters orders in specific regions with an amount in a given range.

**Expected Output:** Orders from North or East between 100 and 500.

**Business Interpretation:** Combines multiple filter conditions for targeted analysis.

## Q10 — Handling NULLs

```
SELECT * FROM customers WHERE region IS NULL;
```

**Explanation:** Finds customer rows where the region field was never filled in.

**Expected Output:** Rows with missing region data.

**Business Interpretation:** Flags data quality issues that need cleaning.

## Q11 — CASE WHEN

```
SELECT order_id, amount,  
CASE WHEN amount > 1000 THEN 'High'  
WHEN amount > 300 THEN 'Medium'  
ELSE 'Low' END AS order_tier  
FROM orders;
```

**Explanation:** Creates a new categorical column based on conditional logic.

**Expected Output:** Each order labeled High, Medium, or Low.

**Business Interpretation:** Segments transactions for easier reporting.

## Q12 — INNER JOIN

```
SELECT o.order_id, c.name, o.amount
FROM orders o
INNER JOIN customers c ON o.customer_id = c.customer_id;
```

**Explanation:** Combines matching rows from orders and customers.

**Expected Output:** Orders with the matching customer name attached.

**Business Interpretation:** Only returns orders that have a valid, matching customer.

## Q13 — LEFT JOIN

```
SELECT c.name, o.order_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id;
```

**Explanation:** Keeps all customers, attaching order data where it exists.

**Expected Output:** Every customer, with NULL order\_id if they never ordered.

**Business Interpretation:** Helps find customers who have never placed an order.

## Q14 — RIGHT and FULL JOIN

```
SELECT * FROM orders o
RIGHT JOIN customers c ON o.customer_id = c.customer_id;
-- FULL JOIN keeps unmatched rows from BOTH tables (not all DBs support it directly)
```

**Explanation:** RIGHT JOIN keeps all rows from the right table; FULL JOIN keeps unmatched rows from both sides.

**Expected Output:** All customers (RIGHT) or all customers and orders (FULL).

**Business Interpretation:** Useful for full data reconciliation between two tables.

## Q15 — SELF JOIN

```
SELECT e1.name AS employee, e2.name AS manager
FROM employees e1
JOIN employees e2 ON e1.manager_id = e2.employee_id;
```

**Explanation:** Joins the employees table to itself to match employees with managers.

**Expected Output:** Employee name paired with their manager's name.

**Business Interpretation:** Builds an organizational reporting structure from one table.

## Q16 — Subquery

```
SELECT name FROM customers
WHERE customer_id IN (
  SELECT customer_id FROM orders WHERE amount > 2000
);
```

**Explanation:** The inner query finds high-value order customer IDs; the outer query gets their names.

**Expected Output:** Names of customers who placed at least one order over 2000.

**Business Interpretation:** Identifies VIP customers for targeted marketing.

## Q17 — Common Table Expression (CTE)

```
WITH regional_sales AS (
  SELECT region, SUM(amount) AS total
  FROM orders o JOIN customers c ON o.customer_id=c.customer_id
  GROUP BY region
)
SELECT * FROM regional_sales WHERE total > 5000;
```

**Explanation:** The CTE pre-calculates regional totals, then the outer query filters them.

**Expected Output:** Regions with total sales above 5000.

**Business Interpretation:** Makes complex, multi-step queries far more readable and reusable.

## Q18 — UNION vs UNION ALL

```
SELECT customer_id FROM orders WHERE region='North'
UNION
SELECT customer_id FROM orders WHERE region='South';
```

**Explanation:** UNION combines results and removes duplicates; UNION ALL keeps duplicates.

**Expected Output:** A unique list of customer IDs from both regions.

**Business Interpretation:** Combines multiple result sets into a single report.

## Q19 — Window Function: ROW\_NUMBER

```
SELECT order_id, customer_id, amount,
ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY amount DESC) AS rn
FROM orders;
```

**Explanation:** Numbers each customer's orders from largest to smallest amount.

**Expected Output:** Row number 1 marks each customer's biggest order.

**Business Interpretation:** Useful for finding each customer's top purchase.

## Q20 — RANK and DENSE\_RANK

```
SELECT customer_id, SUM(amount) AS total,  
RANK() OVER (ORDER BY SUM(amount) DESC) AS rnk  
FROM orders GROUP BY customer_id;
```

**Explanation:** Ranks customers by total spend; RANK() skips numbers after ties, DENSE\_RANK() does not.

**Expected Output:** Customers ranked from highest to lowest spender.

**Business Interpretation:** Builds a customer leaderboard for sales reporting.

## Q21 — LAG and LEAD

```
SELECT order_date, amount,  
LAG(amount) OVER (ORDER BY order_date) AS prev_amount  
FROM orders;
```

**Explanation:** LAG() pulls the previous row's value; LEAD() pulls the next row's value.

**Expected Output:** Each order alongside the previous order's amount.

**Business Interpretation:** Useful for period-over-period comparisons, like month-over-month change.

## Q22 — Date Functions

```
SELECT order_id, EXTRACT(MONTH FROM order_date) AS order_month  
FROM orders;
```

**Explanation:** Extracts the month number from a date column.

**Expected Output:** Each order tagged with its month.

**Business Interpretation:** Enables monthly trend analysis and seasonality checks.

## Q23 — String Functions

```
SELECT UPPER(name) AS name_upper, LENGTH(name) AS name_length  
FROM customers;
```

**Explanation:** Converts text to uppercase and measures its length.

**Expected Output:** Standardized, measured customer names.

**Business Interpretation:** Helps standardize text fields before reporting or matching.

## Q24 — Data Cleaning with TRIM/COALESCE

```
SELECT TRIM(name) AS clean_name, COALESCE(region, 'Unknown') AS region  
FROM customers;
```

**Explanation:** TRIM removes extra spaces; COALESCE replaces NULLs with a default value.

**Expected Output:** Clean names with no missing region labels.

**Business Interpretation:** A common first step before loading data into a report.

## 6.3 More Practical Queries

- `SELECT * FROM products WHERE price BETWEEN 20 AND 50;` — mid-range products.
- `SELECT category, AVG(price) FROM products GROUP BY category;` — average price by category.
- `SELECT COUNT(DISTINCT customer_id) FROM orders;` — unique paying customers.
- `SELECT * FROM orders WHERE order_date >= '2026-01-01';` — orders this year.
- `SELECT department, COUNT(*) FROM employees GROUP BY department ORDER BY COUNT(*) DESC;` — headcount by department.
- `DELETE FROM orders WHERE amount <= 0;` — remove invalid zero/negative orders.
- `UPDATE customers SET region='Unknown' WHERE region IS NULL;` — fill missing region.

## SQL Mini-Project

### Mini-Project: Customer & Sales Report

- Using the customers, orders, and products tables, write a query that returns each customer's name, total number of orders, and total amount spent, sorted from highest to lowest spender.
- Extend it with a CTE that also shows each customer's rank using `RANK()`.
- Add a filter (`HAVING`) to show only customers with 3 or more orders.
- Expected Learning Outcome: comfortably combine `JOIN`, `GROUP BY`, `HAVING`, CTEs, and window functions in a single realistic report.

### Key Takeaways

- `WHERE` filters rows before grouping; `HAVING` filters groups after aggregation.
- `INNER JOIN` keeps only matches; `LEFT JOIN` keeps all rows from the left table.
- CTEs (`WITH ...`) make multi-step queries far more readable than nested subqueries.
- Window functions like `ROW_NUMBER`, `RANK`, `LAG`, and `LEAD` analyze rows in context without collapsing them into groups.

## Practice Questions

- 1 Write a query to find the top 5 customers by total order amount.
- 2 Write a query using `LEFT JOIN` to find customers with zero orders.
- 3 Explain the difference between `RANK()` and `DENSE_RANK()` with an example.
- 4 Write a CTE that calculates average order value per region, then filters regions above the overall average.

## Interview Questions

**Q1. What is the difference between `WHERE` and `HAVING`?**

WHERE filters individual rows before any grouping occurs, while HAVING filters groups after aggregation — for example, filtering groups where SUM(amount) exceeds a threshold.

**Q2. What is the difference between INNER JOIN and LEFT JOIN?**

INNER JOIN returns only rows with matches in both tables, while LEFT JOIN returns all rows from the left table and NULLs for any unmatched right-side columns.

**Q3. What is a CTE and why use one?**

A Common Table Expression, defined with WITH, is a named temporary result set that makes complex queries easier to read and reuse compared to deeply nested subqueries.

**Q4. What is the difference between UNION and UNION ALL?**

UNION removes duplicate rows from the combined result, while UNION ALL keeps every row, including duplicates, and is generally faster.

**Q5. What does the RANK() window function do?**

It assigns a rank to each row within a partition based on an ORDER BY clause, with tied values receiving the same rank and a gap left in the sequence afterward.

**Q6. How do you find duplicate records in SQL?**

Group by the columns that should be unique and use HAVING COUNT(\*) > 1 to surface groups with more than one matching row.

## CHAPTER 7

# Python for Data Analytics

### Learning Objectives

- Understand core Python syntax: variables, data types, and operators.
- Work with lists, tuples, dictionaries, and sets.
- Use conditionals, loops, and functions to process data.
- Read files and perform basic data manipulation in plain Python.

## 7.1 Why Python for Analytics?

Python is a beginner-friendly, general-purpose programming language that has become the most widely used language in data analytics because of its readable syntax and powerful libraries such as Pandas and NumPy, covered in the next two chapters.

## 7.2 Variables and Data Types

```
age = 28 # int
price = 499.99 # float
name = "Data Analyst" # str
is_active = True # bool
```

| Type  | Description     | Example        |
|-------|-----------------|----------------|
| int   | Whole numbers   | 10, -5, 2026   |
| float | Decimal numbers | 3.14, 99.5     |
| str   | Text            | "Sales Report" |
| bool  | True or False   | True, False    |

## 7.3 Lists, Tuples, Dictionaries, and Sets

```
sales = [120, 340, 560, 210] # list: ordered, changeable
coordinates = (28.6, 77.2) # tuple: ordered, unchangeable
customer = {"name": "Riya", "region": "East"} # dictionary: key-value pairs
regions = {"North", "South", "East"} # set: unique, unordered values
```

- **List** — use when order matters and values may change (e.g. daily sales).
- **Tuple** — use for fixed values that should not change (e.g. coordinates).
- **Dictionary** — use for labeled data accessed by key (e.g. a customer record).
- **Set** — use to store unique values and remove duplicates quickly.

## 7.4 Operators

| Category   | Examples                                                                                                                                           |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Arithmetic | <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , <code>//</code> (floor div), <code>%</code> (modulus), <code>**</code> (power) |
| Comparison | <code>==</code> , <code>!=</code> , <code>&gt;</code> , <code>&lt;</code> , <code>&gt;=</code> , <code>&lt;=</code>                                |
| Logical    | <code>and</code> , <code>or</code> , <code>not</code>                                                                                              |

## 7.5 Conditional Statements

```
sales = 750
if sales > 1000:
    tier = "Gold"
elif sales > 500:
    tier = "Silver"
else:
    tier = "Bronze"
print(tier) # Output: Silver
```

## 7.6 Loops

```
sales = [120, 340, 560, 210]
total = 0
for amount in sales:
    total += amount
print(total) # Output: 1230

i = 0
while i < 3:
    print("Row", i)
    i += 1
```

## 7.7 Functions

```
def profit_margin(sales, cost):
    return (sales - cost) / sales * 100

print(profit_margin(1000, 700)) # Output: 30.0
```

## 7.8 List Comprehensions

```
sales = [120, 340, 560, 210]
high_sales = [s for s in sales if s > 300]
print(high_sales) # Output: [340, 560]
```

List comprehensions offer a compact way to build a new list by filtering or transforming an existing one, replacing a longer for-loop.

## 7.9 Exception Handling

```
try:
    result = 100 / 0
except ZeroDivisionError:
    print("Cannot divide by zero")
```

Exception handling prevents a script from crashing when it encounters an unexpected error, such as bad data or a missing file, and lets the analyst handle the problem gracefully.

## 7.10 Reading Files and Working with CSV

```
import csv

with open("sales.csv") as f:
    reader = csv.DictReader(f)
    for row in reader:
        print(row["region"], row["amount"])
```

## 7.11 Basic Data Manipulation

```
sales = [120, 340, 560, 210, 890]
print(sum(sales)) # total
print(sum(sales)/len(sales)) # average
print(max(sales), min(sales))
print(sorted(sales, reverse=True))
```

## Mini-Project: Sales Summary Script

### Mini-Project

- Given a list of dictionaries representing orders (each with 'region' and 'amount' keys), write a Python script that:
  - 1) Calculates total sales per region using a dictionary accumulator.
  - 2) Uses a list comprehension to find all orders above 500.
  - 3) Wraps the file-reading logic in a try/except block.
- Expected Learning Outcome: comfortably combine loops, conditionals, dictionaries, and list comprehensions to summarize real data.

### Key Takeaways

- Lists are mutable and ordered; tuples are immutable; dictionaries store key-value pairs.
- List comprehensions provide a concise way to filter or transform lists.
- try/except prevents scripts from crashing on unexpected errors or bad data.
- Python's built-in functions (sum, max, min, sorted) cover many basic analytics needs.

## Practice Questions

- 1 Write a function that takes a list of sales figures and returns the average.
- 2 Use a list comprehension to extract only even numbers from a list of 1–20.
- 3 Write a dictionary representing a customer record, then print each value using a loop over its keys.
- 4 Write a try/except block that safely handles a missing CSV file.

## Interview Questions

### Q1. What is the difference between a list and a tuple in Python?

Lists are mutable (can be changed after creation) and use square brackets; tuples are immutable and use parentheses, making them suitable for fixed data that should not change.

### Q2. What is a dictionary used for?

It stores data as key-value pairs, allowing fast lookup of a value using its associated key — ideal for representing a single record, like a customer profile.

### Q3. What is a list comprehension?

A concise syntax for creating a new list by applying an expression and optional filter condition to each item in an existing iterable, in a single readable line.

# NumPy

## Learning Objectives

- Understand what NumPy arrays are and why they are faster than lists.
- Create, index, slice, and reshape arrays.
- Perform mathematical and statistical operations on arrays.
- Understand broadcasting for element-wise operations.

## 8.1 What is NumPy?

NumPy (Numerical Python) is a library for fast, memory-efficient numerical computing in Python. Its core object is the **array** (ndarray), which stores elements of the same data type and supports vectorized operations — meaning calculations run on entire arrays at once, without explicit loops.

## 8.2 Creating Arrays

```
import numpy as np

arr = np.array([10, 20, 30, 40])
zeros = np.zeros(5)
ones = np.ones((2, 3))
sequence = np.arange(0, 10, 2) # [0 2 4 6 8]
matrix = np.array([[1, 2], [3, 4]])
```

## 8.3 Indexing and Slicing

```
arr = np.array([10, 20, 30, 40, 50])
print(arr[0]) # 10
print(arr[-1]) # 50
print(arr[1:4]) # [20 30 40]

matrix = np.array([[1, 2, 3], [4, 5, 6]])
print(matrix[1, 2]) # 6 (row 1, column 2)
```

## 8.4 Array Shape and Reshaping

```
arr = np.arange(12)
print(arr.shape) # (12,)
reshaped = arr.reshape(3, 4)
print(reshaped.shape) # (3, 4)
```

**Shape** describes an array's dimensions. **Reshape** changes the shape of an array without changing its data, as long as the total number of elements stays the same.

## 8.5 Mathematical and Statistical Operations

```
sales = np.array([120, 340, 560, 210, 890])
print(sales.sum()) # 2120
print(sales.mean()) # 424.0
print(sales.std()) # standard deviation
print(sales.max(), sales.min())
print(sales * 1.1) # apply a 10% increase to every element
```

## 8.6 Aggregation Across Axes

```
matrix = np.array([[10, 20, 30], [40, 50, 60]])
print(matrix.sum(axis=0)) # column sums: [50 70 90]
print(matrix.sum(axis=1)) # row sums: [60 150]
```

## 8.7 Broadcasting

Broadcasting lets NumPy apply operations between arrays of different shapes without writing explicit loops — for example, adding a single number to every element of an array, or adding a 1-row array to every row of a matrix.

```
prices = np.array([100, 200, 300])
discounted = prices - 20 # subtracts 20 from every element
print(discounted) # [80 180 280]
```

## 8.8 Practical Example

Suppose we have daily sales for a week and want to find which days performed above average:

```

daily_sales = np.array([1200, 950, 1600, 1100, 2000, 1750, 900])
avg = daily_sales.mean()
above_avg_days = daily_sales[daily_sales > avg]
print(avg) # 1357.14
print(above_avg_days) # [1600 2000 1750]

```

| Function                       | Purpose                                     |
|--------------------------------|---------------------------------------------|
| np.array()                     | Create an array from a list or nested list  |
| np.arange(start, stop, step)   | Create a sequence of numbers                |
| array.reshape()                | Change array dimensions                     |
| array.sum() / .mean() / .std() | Aggregate statistics                        |
| array[condition]               | Boolean filtering (e.g. array[array > 100]) |

### Key Takeaways

- NumPy arrays are faster and more memory-efficient than Python lists for numeric data.
- Vectorized operations apply a calculation to an entire array without a manual loop.
- Broadcasting lets arrays of different shapes interact in arithmetic operations.
- Boolean indexing (array[array > value]) is a fast way to filter data.

## Practice Questions

- 1 Create a NumPy array of the numbers 1 to 20 and reshape it into a 4x5 matrix.
- 2 Given an array of daily temperatures, find all days above the average.
- 3 Explain what broadcasting means using a simple example.
- 4 Calculate the row-wise and column-wise sum of a 3x3 matrix.

## Interview Questions

### Q1. Why is NumPy faster than plain Python lists for numerical work?

NumPy arrays store data in contiguous memory blocks of a single type and use vectorized, compiled operations, avoiding the overhead of Python-level loops that lists require.

### Q2. What is broadcasting in NumPy?

It is the mechanism that allows NumPy to perform element-wise operations on arrays of different shapes by automatically expanding the smaller array's dimensions to match the larger one.

## CHAPTER 9

# Pandas

### Learning Objectives

- Understand Series and DataFrame, Pandas' two core structures.
- Load, inspect, filter, and sort data with Pandas.
- Handle missing values, duplicates, and data type conversion.
- Use GroupBy, merge, and pivot tables for real analysis.

## 9.1 Series and DataFrame

A **Series** is a one-dimensional labeled array — essentially a single column. A **DataFrame** is a two-dimensional labeled table made of multiple Series sharing the same index — essentially a spreadsheet inside Python.

```
import pandas as pd

s = pd.Series([100, 200, 300])
df = pd.DataFrame({
    "region": ["North", "South", "East"],
    "sales": [1200, 950, 1600]
})
```

## 9.2 Reading Data

```
df = pd.read_csv("orders.csv")
df_excel = pd.read_excel("orders.xlsx", sheet_name="Sheet1")
```

## 9.3 Inspecting Data

| Method        | Purpose                                       |
|---------------|-----------------------------------------------|
| df.head()     | Preview the first 5 rows                      |
| df.tail()     | Preview the last 5 rows                       |
| df.info()     | Column names, data types, and non-null counts |
| df.describe() | Summary statistics for numeric columns        |
| df.shape      | Number of rows and columns                    |

## 9.4 Selecting and Filtering

```
df["sales"] # select one column
df[["region", "sales"]] # select multiple columns
df[df["sales"] > 1000] # filter rows
df[(df["sales"] > 1000) & (df["region"] == "North")] # multiple conditions
```

## 9.5 Sorting

```
df.sort_values("sales", ascending=False)
```

## 9.6 Handling Missing Values

```
df.isnull().sum() # count missing values per column
df.dropna() # remove rows with any missing value
df.fillna(0) # replace missing values with 0
df["region"].fillna("Unknown", inplace=True)
```

## 9.7 Duplicates and Data Types

```
df.duplicated().sum() # count duplicate rows
df.drop_duplicates(inplace=True)
df["order_date"] = pd.to_datetime(df["order_date"])
df["sales"] = df["sales"].astype(float)
```

## 9.8 Renaming and Applying Functions

```
df.rename(columns={"amt": "sales"}, inplace=True)
df["sales_with_tax"] = df["sales"].apply(lambda x: x * 1.18)
```

## 9.9 GroupBy and Aggregation

```
df.groupby("region")["sales"].sum()
df.groupby("region").agg(
    total_sales=("sales", "sum"),
    avg_sales=("sales", "mean"),
    orders=("order_id", "count")
)
```

GroupBy follows a 'split-apply-combine' pattern: split the data into groups by a column, apply an aggregation to each group, then combine the results into a summary table.

## 9.10 Merge, Join, and Concatenation

```
merged = pd.merge(orders, customers, on="customer_id", how="left")  
combined = pd.concat([df_2025, df_2026]) # stack two DataFrames
```

| how=  | Behavior                                          |
|-------|---------------------------------------------------|
| inner | Keeps only rows with matching keys in both tables |
| left  | Keeps all rows from the left table                |
| right | Keeps all rows from the right table               |
| outer | Keeps all rows from both tables                   |

## 9.11 Pivot Tables in Pandas

```
pd.pivot_table(df, values="sales", index="region",  
columns="category", aggfunc="sum")
```

## 9.12 Date/Time Handling

```
df["order_date"] = pd.to_datetime(df["order_date"])  
df["month"] = df["order_date"].dt.month  
df["weekday"] = df["order_date"].dt.day_name()
```

## Complete Data-Cleaning Example

```
import pandas as pd

df = pd.read_csv("raw_orders.csv")

# 1. Remove duplicates
df.drop_duplicates(inplace=True)

# 2. Fix data types
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
df["sales"] = pd.to_numeric(df["sales"], errors="coerce")

# 3. Handle missing values
df["region"].fillna("Unknown", inplace=True)
df.dropna(subset=["sales"], inplace=True)

# 4. Standardize text
df["region"] = df["region"].str.strip().str.title()

# 5. Save cleaned data
df.to_csv("clean_orders.csv", index=False)
```

### Mini-Project: Sales Data Cleaning & Summary

- Load a raw orders CSV with missing values, duplicates, and inconsistent text.
- Clean the dataset following the five steps shown above.
- Use groupby to calculate total and average sales per region and category.
- Merge in a customers table to add customer names to the summary.
- Expected Learning Outcome: perform an end-to-end clean-and-summarize workflow using only Pandas.

### Key Takeaways

- A DataFrame is Pandas' core table structure, built from labeled Series columns.
- groupby() follows split-apply-combine to summarize data by category.
- merge() combines tables on a key column, similar to a SQL JOIN.
- Always check dtypes and missing values early with info() and isnull().sum().

## Practice Questions

- 1 Load a CSV and print the first 5 rows, then check for missing values.
- 2 Filter a DataFrame to show only rows where sales exceed the column average.
- 3 Group an orders DataFrame by category and calculate total and average sales.

- 4 Merge two DataFrames (orders and customers) using a left join on customer\_id.

## Interview Questions

### Q1. What is the difference between a Series and a DataFrame?

A Series is a single labeled column of data, while a DataFrame is a two-dimensional table made up of multiple Series sharing the same index.

### Q2. How do you handle missing values in Pandas?

Detect them with `isnull().sum()`, then either remove them with `dropna()` or replace them with a sensible value using `fillna()`, depending on the situation.

### Q3. What does `groupby()` do in Pandas?

It splits the DataFrame into groups based on a column's values, applies an aggregation function to each group, and combines the results into a summary — the split-apply-combine pattern.

### Q4. What is the difference between `merge()` and `concat()`?

`merge()` combines DataFrames based on a common key column, similar to a SQL join, while `concat()` simply stacks DataFrames on top of or beside each other.

# Data Cleaning and Preprocessing

## Learning Objectives

- Understand why data cleaning is the foundation of trustworthy analysis.
- Identify and fix missing data, duplicates, and inconsistent formatting.
- Understand standardization, normalization, and outlier handling.
- Apply a before-and-after cleaning example to a realistic dataset.

## 10.1 Why Data Cleaning Matters

Real-world data is almost never analysis-ready. It arrives with missing entries, duplicate records, inconsistent spelling, and incorrect data types. Analyzing dirty data produces misleading results — the saying 'garbage in, garbage out' captures why cleaning typically takes up the largest share of an analyst's time.

## 10.2 Common Data Quality Issues

| Issue                   | Example                           | Typical Fix                                |
|-------------------------|-----------------------------------|--------------------------------------------|
| Missing data            | Blank 'Region' field              | Fill with a default or drop the row        |
| Duplicate records       | Same order entered twice          | Remove duplicates on key columns           |
| Incorrect data types    | Sales stored as text              | Convert to numeric type                    |
| Invalid values          | Age = -5, Discount = 150%         | Flag and correct or remove                 |
| Inconsistent formatting | 'ny', 'NY', 'New York'            | Standardize to one format                  |
| Outliers                | Order of \$500,000 in retail data | Investigate before deciding to keep/remove |

## 10.3 Standardization vs Normalization

**Standardization** rescales data to have a mean of 0 and a standard deviation of 1, useful when comparing variables on different scales. **Normalization** rescales data into a fixed range, typically 0 to 1, often used before feeding data into distance-based algorithms.

## 10.4 Text and Date Cleaning

- Trim leading/trailing spaces and fix inconsistent capitalization.
- Standardize category labels (e.g. 'M', 'Male', 'male' → 'Male').
- Convert date strings into a proper date type, and standardize the format (e.g. DD/MM/YYYY vs MM/DD/YYYY) before analysis.
- Remove or fix impossible dates, such as a signup date in the future.

## 10.5 Handling Categorical Data

Categorical columns should be checked for spelling inconsistencies and unexpected categories. Grouping rare categories into an 'Other' bucket can simplify analysis when a category has very few observations.

## 10.6 Validation Rules

- Numeric fields should fall within a logical range (e.g. discount 0–100%).
- Foreign key values (like customer\_id in an orders table) should exist in the related table.
- Dates should follow a logical order (e.g. ship\_date should not precede order\_date).

## 10.7 Before-and-After Example

| Column        | Before (Raw)                | After (Cleaned)             |
|---------------|-----------------------------|-----------------------------|
| customer_name | "riya SHAH "                | "Riya Shah"                 |
| region        | "ny", "New York", "NY "     | "New York" (standardized)   |
| sales         | "\$1,200" (text)            | 1200.0 (numeric)            |
| order_date    | "03-14-2026" / "2026/03/14" | 2026-03-14 (datetime)       |
| discount      | 150 (invalid %)             | flagged and corrected to 15 |
| duplicate row | Order #A102 appears twice   | duplicate removed           |

### Why This Matters

If the raw 'sales' column above were analyzed as text, a SUM formula or Pandas aggregation would fail or silently return 0 — only after cleaning does the data become trustworthy for reporting or dashboards.

### Key Takeaways

- Data cleaning typically takes more time than the analysis itself, but is essential.
- Common issues are missing data, duplicates, wrong types, invalid values, and inconsistent formatting.
- Validation rules catch logical errors that basic type-checking would miss.
- Always inspect a before-and-after sample to confirm cleaning worked correctly.

## Practice Questions

- 1 List five common data quality issues and one fix for each.
- 2 Explain the difference between standardization and normalization.
- 3 Given the region values 'ny', 'NY', 'New York', explain how you would clean this column.
- 4 Why is it dangerous to remove all outliers automatically without investigation?

## Interview Questions

**Q1. What steps do you take to clean a new dataset?**

I first inspect data types and missing values, remove duplicates, standardize text and categorical fields, convert dates and numbers to the correct type, and validate values against logical business rules.

**Q2. How do you decide whether to remove or investigate an outlier?**

I check whether the value is a genuine data entry error (e.g. an impossible negative age) versus a real but extreme business event (e.g. a legitimately large order) — only the former should typically be removed.

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# Exploratory Data Analysis (EDA)

## Learning Objectives

- Understand what EDA is and why it precedes formal analysis.
- Distinguish univariate, bivariate, and multivariate analysis.
- Choose the right chart for distribution, comparison, and relationship questions.
- Perform EDA in Python using Pandas and Matplotlib.

## 11.1 What is EDA?

Exploratory Data Analysis (EDA) is the process of investigating a dataset to summarize its main characteristics, often using visual methods, before applying formal modeling or reporting. EDA helps analysts understand distributions, spot anomalies, and form hypotheses worth testing further.

## 11.2 Why EDA Matters

- Reveals data quality issues that cleaning may have missed.
- Shows the shape of each variable's distribution (skewed, normal, bimodal).
- Uncovers relationships between variables before formal statistical testing.
- Prevents analysts from applying the wrong technique to the wrong data.

## 11.3 Univariate, Bivariate, and Multivariate Analysis

| Type         | Definition                                           | Example                       |
|--------------|------------------------------------------------------|-------------------------------|
| Univariate   | Examines one variable at a time                      | Distribution of order amounts |
| Bivariate    | Examines the relationship between two variables      | Sales vs advertising spend    |
| Multivariate | Examines relationships among three or more variables | Sales vs region vs season     |

## 11.4 Distribution, Outlier, and Correlation Analysis

**Distribution analysis** studies how values are spread (using histograms or box plots). **Outlier detection** flags unusual values for investigation. **Correlation analysis** studies the strength of linear relationships between numeric variables, often visualized with a heatmap of correlation coefficients.

## 11.5 Choosing the Right Chart

| Chart     | Best For                           |
|-----------|------------------------------------|
| Bar chart | Comparing totals across categories |

| Chart        | Best For                                                 |
|--------------|----------------------------------------------------------|
| Line chart   | Trends over time                                         |
| Histogram    | Distribution of a single numeric variable                |
| Box plot     | Spread, median, and outliers of a numeric variable       |
| Scatter plot | Relationship between two numeric variables               |
| Pie chart    | Proportional share of a whole (best with few categories) |
| Heatmap      | Correlation strength between many variables at once      |

## 11.6 EDA in Python: Practical Example

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("orders.csv")

# Univariate: distribution of sales
df["sales"].hist(bins=20)
plt.title("Distribution of Order Sales")
plt.show()

# Bivariate: sales vs discount
plt.scatter(df["discount"], df["sales"])
plt.xlabel("Discount"); plt.ylabel("Sales")
plt.show()

# Correlation heatmap data
print(df.corr(numeric_only=True))

# Outlier check with a box plot
df.boxplot(column="sales")
plt.show()
```

## 11.7 Pattern and Trend Identification

### Applied Example

Plotting monthly sales for an e-commerce store as a line chart reveals a consistent spike every November — a seasonal pattern tied to holiday shopping that would be missed by looking at raw numbers alone.

### Key Takeaways

- EDA comes before formal modeling and helps form hypotheses, not conclusions.
- Univariate analysis studies one variable; bivariate studies relationships between two; multivariate studies three or more together.
- Histograms and box plots reveal distribution shape and outliers.
- Scatter plots and correlation matrices reveal relationships between numeric variables.

## Practice Questions

- 1 Given a sales dataset, which chart would you use to check for a monthly seasonal trend? Why?
- 2 Explain the difference between univariate and bivariate analysis with examples.
- 3 How would you visually check for outliers in a numeric column?
- 4 Write Pandas/Matplotlib code to plot a histogram of a 'price' column.

## Interview Questions

### Q1. What is the goal of exploratory data analysis?

To understand a dataset's structure, distributions, and relationships before applying formal statistical tests or building models — essentially forming hypotheses and catching issues early.

### Q2. Which chart would you use to detect outliers in a numeric column?

A box plot, since it visually displays the median, quartiles, and any points beyond 1.5 times the interquartile range as potential outliers.

### Q3. What does a correlation heatmap show?

It displays the pairwise correlation coefficients between numeric variables as a color-coded grid, making it easy to spot strong relationships across many variables at once.

# Data Visualization

### Learning Objectives

- Learn the principles of effective, honest data visualization.
- Match chart types to the story the data needs to tell.
- Understand dashboard design and color-use best practices.
- Avoid common mistakes that mislead viewers.

## 12.1 Principles of Effective Visualization

- Choose the simplest chart that clearly answers the question.
- Every chart should have one clear message, not a data dump.
- Label axes, units, and sources so the chart stands on its own.
- Reduce clutter — remove gridlines, borders, and decorations that don't add meaning.
- Use color purposefully, not decoratively.

## 12.2 Choosing the Correct Chart

| Goal                                  | Recommended Chart                                  |
|---------------------------------------|----------------------------------------------------|
| Compare categories                    | Bar chart                                          |
| Show change over time                 | Line chart or area chart                           |
| Show parts of a whole                 | Pie chart (only for few categories) or stacked bar |
| Show relationship between two numbers | Scatter plot                                       |
| Show distribution                     | Histogram or box plot                              |
| Show many correlations at once        | Heatmap                                            |
| Highlight a single key number         | KPI card                                           |

## 12.3 Dashboard Design

A good dashboard answers a specific set of business questions at a glance. Effective dashboards place the most important KPIs at the top left (where the eye naturally lands first), group related charts together, and allow filtering (via slicers) without cluttering the layout.

## 12.4 Color Selection

- Use a small, consistent color palette across all charts in a report.
- Reserve a distinct color (like orange or red) to highlight what matters most.

- Use color to reinforce meaning — e.g. green for positive, red for negative — consistently throughout the entire dashboard.
- Consider color-blind-friendly palettes for wide audiences.

## 12.5 Avoiding Misleading Charts

| Mistake                           | Why It Misleads                                     |
|-----------------------------------|-----------------------------------------------------|
| Truncated y-axis                  | Makes small differences look dramatic               |
| 3D charts                         | Distort proportions and are hard to read accurately |
| Too many colors/categories        | Overwhelms the viewer, obscures the message         |
| Inconsistent scales across charts | Makes side-by-side comparison inaccurate            |
| Pie chart with many slices        | Small slices become impossible to compare           |

### Example of a Misleading Chart

A bar chart with a y-axis starting at 90 (instead of 0) can make a change from 92 to 95 look like a massive jump, when it is really only a 3% increase — always check whether an axis has been truncated before trusting a chart's visual impression.

## 12.6 Storytelling with Data

The strongest analytics deliverables tell a story: start with the business question, show the key finding first, support it with one or two supporting charts, and end with a clear, specific recommendation. Raw dashboards inform; a data story persuades and drives action.

### Key Takeaways

- Choose charts based on the question being answered, not personal preference.
- Simplicity and clarity beat decoration in professional visualization.
- Truncated axes and 3D effects are common ways charts mislead viewers.
- A strong data story leads with the finding, not just the data.

## Practice Questions

- 1 Which chart type would best show monthly revenue trends over two years?
- 2 Explain why a pie chart with 12 slices is usually a poor choice.
- 3 Describe two ways a chart's y-axis can be used to mislead viewers.
- 4 Design (in words) a one-page sales dashboard layout with three KPIs and two charts.

## Interview Questions

### Q1. How do you decide which chart type to use?

I start from the question being asked — comparison, trend, distribution, or relationship — and pick the simplest chart type that answers it clearly, avoiding decorative chart types that don't add value.

## **Q2. What makes a dashboard effective?**

It focuses on a small number of key metrics relevant to its audience, uses consistent colors and scales, and is organized so the most important information is immediately visible.

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# Power BI

## Learning Objectives

- Understand the Power BI ecosystem and the report-building workflow.
- Import, clean, and model data using Power Query and relationships.
- Write core DAX measures and calculated columns.
- Build and publish an interactive dashboard.

## 13.1 What is Power BI?

Power BI is Microsoft's business intelligence platform for connecting to data, transforming it, modeling relationships, and building interactive reports and dashboards. It is widely used because it integrates closely with Excel and other Microsoft tools.

## 13.2 The Power BI Ecosystem

| Component        | Purpose                                                |
|------------------|--------------------------------------------------------|
| Power BI Desktop | Free application for building reports on your computer |
| Power BI Service | Cloud platform for publishing and sharing reports      |
| Power BI Mobile  | App for viewing dashboards on phones/tablets           |
| Power Query      | Tool for importing and transforming data               |
| DAX              | Formula language for measures and calculated columns   |

## 13.3 Importing and Transforming Data

Data is imported via 'Get Data' from sources such as Excel, SQL Server, or web APIs. **Power Query** is then used to clean and reshape the data before it reaches the report — removing columns, filtering rows, fixing data types, splitting columns, and merging queries, all without writing code.

## 13.4 Data Modeling and Relationships

Power BI works best with multiple related tables (a **star schema**): one central 'fact' table (e.g. Sales) connected to smaller 'dimension' tables (e.g. Customers, Products, Dates) via **relationships**, similar to primary/foreign keys in SQL. This keeps the model efficient and flexible for slicing data many ways.

## 13.5 Measures vs Calculated Columns

|            | Measure                                | Calculated Column                    |
|------------|----------------------------------------|--------------------------------------|
| Calculated | At query time, based on filter context | At data refresh, stored in the table |

|          | Measure                                 | Calculated Column                           |
|----------|-----------------------------------------|---------------------------------------------|
| Used for | Aggregations (sums, ratios, YoY change) | Row-level classification (e.g. tier labels) |
| Example  | Total Sales = SUM(Sales[Amount])        | Tier = IF(Sales[Amount]>1000,"High","Low")  |

## 13.6 Introduction to DAX

DAX (Data Analysis Expressions) is the formula language used to build measures and calculated columns in Power BI, similar in spirit to Excel formulas but designed for relational, multi-table models.

### Common DAX Functions

```
-- SUM
Total Sales = SUM(Sales[Amount])

-- COUNT
Order Count = COUNT(Sales[OrderID])

-- DISTINCTCOUNT
Unique Customers = DISTINCTCOUNT(Sales[CustomerID])

-- CALCULATE
North Sales = CALCULATE(SUM(Sales[Amount]), Sales[Region]="North")

-- FILTER
High Value Orders = FILTER(Sales, Sales[Amount] > 1000)

-- IF
Tier = IF(Sales[Amount] > 1000, "High", "Low")
```

## 13.7 Date Functions and Time Intelligence

```
-- Year-to-date sales
YTD Sales = TOTALYTD(SUM(Sales[Amount]), 'Date'[Date])

-- Sales from the same period last year
Sales LY = CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR('Date'[Date]))
```

Time intelligence functions require a proper 'Date' table marked as the model's date table, connected to the fact table by date.

## 13.8 Visualizations, Slicers, and Drill-Down

- **Visualizations** — bar, line, map, card, and matrix visuals built by dragging fields onto the canvas.
- **Slicers** — on-report filter controls (e.g. a Region dropdown).
- **Filters pane** — page-level or report-level filtering, hidden from the main canvas.
- **Drill-down** — clicking into a hierarchy (Year → Quarter → Month) for more detail.

## 13.9 Publishing Reports

Once built, a report is published to the Power BI Service using 'Publish', where it can be organized into workspaces, shared with colleagues, and scheduled to refresh automatically from the underlying data source.

## 13.10 Power BI Best Practices

- Build a star schema instead of one giant flat table.
- Use measures (not calculated columns) for aggregations whenever possible.
- Hide technical/helper columns from the report view.
- Keep a consistent color theme and limit visuals per page.
- Name measures and tables clearly for report maintainability.

## Power BI Dashboard Project

### Mini-Project: Sales Performance Dashboard

- Import a Sales fact table and Customers/Products/Date dimension tables.
- Build relationships between the fact and dimension tables.
- Create measures: Total Sales, Total Profit, Profit Margin %, YTD Sales.
- Build visuals: a KPI card row, a sales trend line chart, a bar chart by category, and a map by region.
- Add slicers for Region and Date, then publish the report.
- Expected Learning Outcome: build a complete, interactive BI dashboard from raw tables to a published report.

### Key Takeaways

- Power Query handles data import/cleaning; DAX handles calculations.
- A star schema (fact + dimension tables) keeps models efficient and flexible.
- Measures calculate dynamically based on filter context; calculated columns are fixed at refresh time.
- CALCULATE is one of the most powerful and commonly used DAX functions.

## Practice Questions

- 1 Explain the difference between a measure and a calculated column with an example.
- 2 Write a DAX measure to calculate total sales for the 'East' region only.
- 3 What is a star schema, and why is it preferred over one flat table?
- 4 Describe the steps to publish a Power BI report and share it with a team.

## Interview Questions

### Q1. What is the difference between Power Query and DAX?

Power Query is used to import and transform (shape) data before it enters the model, while DAX is used to write calculations — measures and calculated columns — on data already inside the model.

### Q2. What is CALCULATE used for in DAX?

It evaluates an expression under a modified filter context, making it one of the most powerful DAX functions for building conditional aggregations like sales for a specific region or period.

### Q3. What is a star schema?

A data modeling approach with one central fact table (containing transactional measures) connected to multiple dimension tables (containing descriptive attributes), which improves performance and makes filtering and relationships simpler.

### Q4. What is the difference between a measure and a calculated column?

A measure is calculated dynamically at query time based on the current filter context, while a calculated column is computed once at data refresh and stored as a static value in the table.

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## CHAPTER 14

# Tableau

### Learning Objectives

- Understand Tableau's interface and core building blocks.
- Distinguish dimensions from measures.
- Build filters, calculated fields, and parameters.
- Combine charts into an interactive dashboard and story.

## 14.1 What is Tableau?

Tableau is a leading data visualization platform known for its drag-and-drop interface, which lets analysts build rich, interactive visuals quickly without writing code. It connects to a wide range of data sources, from spreadsheets to cloud data warehouses.

## 14.2 The Tableau Interface

- **Data pane** — lists all fields available from the connected data source.
- **Shelves** (Columns/Rows) — where fields are dragged to build a view.
- **Marks card** — controls color, size, label, and detail of the visual.
- **Sheets, Dashboards, Stories** — individual charts, combined dashboards, and a guided narrative sequence, respectively.

## 14.3 Dimensions and Measures

| Concept   | Definition                                          | Example                         |
|-----------|-----------------------------------------------------|---------------------------------|
| Dimension | Qualitative, categorical fields used to slice data  | Region, Category, Customer Name |
| Measure   | Quantitative, numeric fields that can be aggregated | Sales, Profit, Quantity         |

Tableau automatically classifies each field as a dimension (usually blue) or a measure (usually green) when data is connected, though this classification can be changed manually if needed.

## 14.4 Filters

Filters restrict which data appears in a view. They can be applied to dimensions (e.g. show only 'North' and 'East'), measures (e.g. Sales > 500), or dates (e.g. last 12 months), and can be added to a dashboard as an interactive control.

## 14.5 Calculated Fields

Calculated fields let analysts create new fields using formulas — similar to Excel formulas or DAX measures. Example:

### Calculated Field Example

Profit Margin =  $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$  — this creates a new measure that can be formatted as a percentage and used in any chart.

## 14.6 Parameters

Parameters are user-controlled input values (like a dropdown or slider) that can dynamically change a calculated field, a filter, or a reference line — for example, letting a viewer choose which metric (Sales or Profit) a chart displays.

## 14.7 Charts in Tableau

| Chart        | Built By                                            |
|--------------|-----------------------------------------------------|
| Bar chart    | Dimension on Rows, Measure on Columns               |
| Line chart   | Date on Columns, Measure on Rows                    |
| Map          | Geographic field (e.g. Country/State) auto-detected |
| Scatter plot | Two measures placed on Columns and Rows             |
| Heatmap      | Two dimensions with color/size mapped to a measure  |

## 14.8 Dashboards and Stories

A **Dashboard** combines multiple sheets into a single interactive view, often with shared filters ('actions') that let clicking one chart filter the others. A **Story** arranges dashboards or sheets into a sequence of slides that walk viewers through a data narrative.

## 14.9 Dashboard Best Practices

- Keep 3–5 visuals per dashboard to avoid clutter.
- Use consistent color encoding for the same measure across sheets.
- Add dashboard actions (filter/highlight) instead of duplicating filters.
- Design for the target screen size (desktop vs mobile layout).

## Practical Tableau Example

### Mini-Project: Regional Performance Dashboard

- Connect to a Sales dataset with Region, Category, Sales, and Profit.
- Build a bar chart of Sales by Category and a map of Sales by Region.
- Create a calculated field for Profit Margin and add it as a color encoding.
- Combine both sheets into a dashboard with a Region filter action.
- Expected Learning Outcome: build a functioning, interactive Tableau dashboard from a raw dataset.

### Key Takeaways

- Dimensions are categorical (blue); measures are numeric and aggregatable (green).
- Calculated fields extend Tableau with custom formulas, similar to Excel/DAX.
- Parameters allow viewers to dynamically control what a chart displays.
- Dashboards combine sheets with shared interactivity via actions.

## Interview Questions

### Q1. What is the difference between a dimension and a measure in Tableau?

Dimensions are qualitative fields used to categorize and slice data (like Region or Category), while measures are quantitative fields that can be aggregated, such as Sales or Profit.

### Q2. What is a calculated field used for?

It lets you define a new field using a formula based on existing fields — for example, calculating profit margin as Profit divided by Sales — for use in any chart.

### Q3. What is the difference between a dashboard and a story in Tableau?

A dashboard combines multiple sheets into a single interactive view, while a story arranges dashboards or sheets into a sequential, presentation-style narrative.

# Business Intelligence and KPI Analysis

## Learning Objectives

- Understand the relationship between BI, KPIs, metrics, dimensions and measures.
- Learn standard KPIs across sales, marketing, finance, customer, and HR.
- Calculate common e-commerce KPIs from raw data.
- Translate KPI trends into business decisions.

## 15.1 Business Intelligence, KPIs, and Metrics

**Business Intelligence** is the practice of monitoring business performance through structured reporting. A **KPI** (Key Performance Indicator) is a specific, important metric tied to a business goal. A **metric** is any measurable value; not every metric is a KPI — only the ones an organization tracks to judge success. **Dimensions** describe context (Region, Product); **measures** are the numeric values being tracked (Sales, Profit).

## 15.2 KPIs by Department

| Department | Example KPIs                                                |
|------------|-------------------------------------------------------------|
| Sales      | Revenue, Sales Growth %, Win Rate, Average Deal Size        |
| Marketing  | Conversion Rate, Cost per Lead, Return on Ad Spend          |
| Finance    | Gross Margin, Operating Expenses, Net Profit Margin         |
| Customer   | Retention Rate, Churn Rate, Customer Lifetime Value         |
| HR         | Employee Turnover Rate, Time-to-Hire, Employee Satisfaction |
| E-commerce | Conversion Rate, Average Order Value, Cart Abandonment Rate |

## 15.3 Key Formulas

### Formula

Conversion Rate =  $(\text{Conversions} \div \text{Total Visitors}) \times 100$

### Formula

Customer Acquisition Cost (CAC) =  $\text{Total Marketing \& Sales Cost} \div \text{New Customers Acquired}$

### Formula

Customer Lifetime Value (CLV) =  $\text{Average Order Value} \times \text{Purchase Frequency} \times \text{Customer Lifespan}$

**Formula**

Retention Rate =  $((\text{Customers at End} - \text{New Customers}) \div \text{Customers at Start}) \times 100$

**Formula**

Churn Rate =  $(\text{Customers Lost During Period} \div \text{Customers at Start of Period}) \times 100$

**Formula**

Average Order Value (AOV) =  $\text{Total Revenue} \div \text{Number of Orders}$

## 15.4 Worked Example

**Numerical Example**

An online store had 10,000 visitors and 250 purchases last month, generating \$30,000 in revenue. Conversion Rate =  $(250 \div 10,000) \times 100 = 2.5\%$ . Average Order Value =  $\$30,000 \div 250 = \$120$ .

## 15.5 Turning KPIs into Decisions

KPIs are only useful when tied to action. A falling conversion rate might trigger a website usability review; a rising CAC relative to CLV might trigger a pause on a specific ad channel; a rising churn rate might trigger a customer retention campaign. Analysts should always pair a KPI trend with a recommended next step.

**Key Takeaways**

- A KPI is a metric tied directly to a strategic business goal.
- CAC and CLV should always be viewed together — a healthy business needs CLV to exceed CAC by a meaningful margin.
- Retention and churn rates are two sides of the same customer-loyalty coin.
- Every KPI report should end with a recommended action, not just a number.

## Practice Questions

- 1 Calculate the conversion rate for a website with 50,000 visitors and 1,200 sales.
- 2 A company spends \$20,000 on marketing and acquires 400 new customers. Calculate the CAC.
- 3 Explain why CLV should be compared against CAC rather than viewed alone.
- 4 List three KPIs relevant to a subscription-based business and explain why.

## Interview Questions

**Q1. What is the difference between a metric and a KPI?**

A metric is any measurable value, while a KPI is a specific metric an organization has chosen to track closely because it directly reflects progress toward a strategic goal.

**Q2. How do you calculate Customer Lifetime Value?**

By multiplying average order value, purchase frequency, and average customer lifespan, giving an estimate of total revenue expected from a typical customer over their relationship with the business.

**Q3. Why is churn rate an important KPI?**

It measures the pace at which a business is losing customers, which directly affects revenue stability and signals whether retention efforts are working.

DATA-ANALYTICS-HUB

# Real-World Project: E-Commerce Sales Analytics

### Learning Objectives

- Apply everything learned so far to one realistic, end-to-end project.
- Practice the full workflow: SQL, Excel, Python, EDA, visualization, and Power BI.
- Turn analysis into concrete business recommendations.

## 16.1 Business Problem

'ShopEase', a mid-sized online retailer, wants to understand which regions, categories, and discount levels are driving — or hurting — profitability, and wants a dashboard the leadership team can use every week.

## 16.2 The Dataset

| Column             | Description                         |
|--------------------|-------------------------------------|
| Order ID           | Unique identifier for each order    |
| Order Date         | Date the order was placed           |
| Customer ID / Name | Identifies the purchasing customer  |
| Product / Category | What was purchased and its category |
| Region             | Geographic region of the customer   |
| Quantity           | Units purchased in the order        |
| Sales              | Revenue generated by the order      |
| Discount           | Discount percentage applied         |
| Profit             | Profit earned on the order          |

## 16.3 Step 1–2: Business Understanding and Data Collection

The dataset is exported from the company's order database into a CSV file covering the last two years of transactions.

## 16.4 Step 3–4: Data Understanding and Cleaning

```
import pandas as pd

df = pd.read_csv("shopease_orders.csv")
df.info()
df.drop_duplicates(inplace=True)
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Region"] = df["Region"].str.strip().str.title()
df.dropna(subset=["Sales", "Profit"], inplace=True)
```

## 16.5 Step 5: SQL Analysis

```
SELECT category, region, SUM(sales) AS total_sales, SUM(profit) AS total_profit
FROM orders
GROUP BY category, region
ORDER BY total_profit ASC;
```

This query immediately surfaces which category/region combinations are least profitable — a key input for the discount policy decision.

## 16.6 Step 6: Excel Analysis

A Pivot Table summarizes Profit by Discount band (0–10%, 10–20%, 20–30%, 30%+), using a helper column with a nested IF formula to bucket each order's discount level.

```
=IF(Discount<=0.1, "0-10%", IF(Discount<=0.2, "10-20%",
IF(Discount<=0.3, "20-30%", "30%+")))
```

## 16.7 Step 7: Python Analysis and EDA

```
profit_by_discount = df.groupby(
pd.cut(df["Discount"], bins=[0,0.1,0.2,0.3,1])
)["Profit"].mean()
print(profit_by_discount)
```

### Key Finding

The analysis shows average profit turns negative once discounts exceed 30%, particularly in the Furniture category — discounts beyond this point are effectively selling at a loss.

## 16.8 Step 8–9: Visualization and Power BI Dashboard

- A bar chart of Profit by Discount Band highlights the loss-making tier.

- A map visual shows Profit by Region.
- A line chart shows the monthly Sales and Profit trend.
- KPI cards display Total Sales, Total Profit, and overall Profit Margin.
- A Region and Category slicer lets leadership drill into any segment.

## 16.9 Step 10–11: KPI Calculation

| KPI                        | Result                     |
|----------------------------|----------------------------|
| Total Sales                | \$1,842,300                |
| Total Profit               | \$187,650                  |
| Overall Profit Margin      | 10.2%                      |
| Average Order Value        | \$134                      |
| Orders with Discount > 30% | 1,120 orders (9% of total) |

## 16.10 Step 12–13: Business Insights and Recommendations

- 1 Cap discounts at 25% for the Furniture category, where margins are thinnest.
- 2 Investigate the South region's below-average profit margin despite strong sales.
- 3 Reinforce the November sales spike with adequate inventory planning.
- 4 Introduce a monthly dashboard review so pricing decisions use current data.

### Key Takeaways

- A real project moves through SQL, Excel, Python, EDA, and BI tools in sequence, each playing to its strength.
- The final deliverable is not a chart — it is a specific, actionable recommendation.
- Discount policy is a common, high-impact finding in retail profitability analysis.

### Extend This Project

- Re-run the discount-band analysis split by Region as well as Category.
- Build a what-if slicer in Power BI to simulate capping discounts at 25%.
- Write a one-page executive summary of the findings and recommendations.

# Second Project: Customer Analytics

## Learning Objectives

- Analyze customer segments, purchase behavior, and retention.
- Calculate repeat-purchase and churn metrics from transaction data.
- Estimate customer lifetime value and identify at-risk customers.

## 17.1 Business Problem

ShopEase's leadership wants to understand who its most valuable customers are, how often customers return, and which customers are at risk of churning, so the marketing team can run a targeted retention campaign.

## 17.2 Customer Segmentation (RFM Analysis)

A common technique is **RFM analysis**, which scores each customer on:

| Dimension     | Meaning                                  |
|---------------|------------------------------------------|
| Recency (R)   | How recently the customer last purchased |
| Frequency (F) | How often the customer purchases         |
| Monetary (M)  | How much the customer has spent in total |

Customers scoring high on all three (recent, frequent, high-spending) are the most valuable segment and are strong candidates for loyalty programs; customers low on Recency but historically high on Frequency and Monetary are prime candidates for a win-back campaign.

## 17.3 Calculating RFM in Python

```
import pandas as pd

snapshot_date = df["Order Date"].max()

rfm = df.groupby("Customer ID").agg(
    Recency=("Order Date", lambda x: (snapshot_date - x.max()).days),
    Frequency=("Order ID", "count"),
    Monetary=("Sales", "sum")
)

rfm["R_score"] = pd.qcut(rfm["Recency"], 4, labels=[4,3,2,1])
rfm["F_score"] = pd.qcut(rfm["Frequency"].rank(method="first"), 4, labels=[1,2,3,4])
rfm["M_score"] = pd.qcut(rfm["Monetary"], 4, labels=[1,2,3,4])
```

## 17.4 Purchase Frequency and Repeat Customers

### Formula

Repeat Purchase Rate = (Customers with 2+ Orders ÷ Total Customers) × 100

### Numerical Example

Out of 5,000 total customers, 1,450 placed more than one order. Repeat Purchase Rate =  $(1,450 \div 5,000) \times 100 = 29\%$ .

## 17.5 Revenue per Customer

```
revenue_per_customer = df.groupby("Customer ID")["Sales"].sum()
print(revenue_per_customer.mean()) # average revenue per customer
print(revenue_per_customer.sort_values(ascending=False).head(10)) # top 10
```

## 17.6 Customer Retention and Churn

A customer is often defined as 'churned' if they have not purchased within a defined window (e.g. 90 days for a fast-moving retailer). Tracking churn month over month reveals whether retention efforts are working.

```
last_purchase = df.groupby("Customer ID")["Order Date"].max()
days_since = (snapshot_date - last_purchase).dt.days
churned = days_since > 90
print(f"Churn rate: {churned.mean()*100:.1f}%")
```

## 17.7 Customer Lifetime Value

Using the CLV formula from Chapter 15 (Average Order Value × Purchase Frequency × Customer Lifespan), analysts can estimate which customer segments are worth the most marketing investment.

## 17.8 The Complete Analytical Process

- 1 Aggregate raw orders into one row per customer.
- 2 Calculate Recency, Frequency, and Monetary scores (RFM).
- 3 Segment customers into groups (e.g. Champions, At-Risk, Lost).
- 4 Calculate repeat purchase rate, churn rate, and CLV per segment.
- 5 Recommend targeted actions per segment (loyalty reward vs win-back offer).

### Key Takeaways

- RFM analysis is a simple, powerful way to segment customers without complex modeling.
- Repeat purchase rate and churn rate are two key views of customer loyalty.
- CLV should guide how much a business is willing to spend to retain a customer.
- Segment-specific recommendations are more actionable than one-size-fits-all advice.

### Mini-Project: Customer Segmentation

- Using a transactions dataset, calculate RFM scores for every customer.
- Segment customers into at least three groups based on their RFM scores.
- Calculate the churn rate and average CLV for each segment.
- Recommend one specific marketing action for each segment.
- Expected Learning Outcome: turn raw transaction data into a customer segmentation strategy end to end.

DATA-ANALYTICS-HUB

# Data Analyst Interview Preparation

### Learning Objectives

- Review high-frequency interview questions across every major topic.
- Practice concise, confident answers suitable for real interviews.
- Build a mental checklist of core concepts before interview day.

## 18.1 Basic / General Questions

(20 questions)

### Q1. What is data analytics?

The process of examining raw data to draw conclusions and support business decisions, using tools, statistics, and domain knowledge.

### Q2. What is the data analytics lifecycle?

Business understanding, data collection, cleaning, transformation, exploration, analysis, visualization, communication, and decision-making.

### Q3. What is the difference between data and information?

Data is raw, unprocessed facts; information is data that has been organized and interpreted to be meaningful.

### Q4. What is the difference between structured and unstructured data?

Structured data fits a fixed schema of rows and columns; unstructured data (images, text) has no predefined organization.

### Q5. What is a KPI?

A Key Performance Indicator — a specific, measurable metric tied directly to a strategic business goal.

### Q6. What is the difference between qualitative and quantitative data?

Qualitative data is descriptive/categorical; quantitative data is numeric and measurable.

### Q7. What does a typical data analyst workflow look like?

Understand the problem, collect and clean data, explore it, analyze it, visualize findings, and present recommendations.

### Q8. What is the difference between primary and secondary data?

Primary data is collected first-hand for a specific purpose; secondary data is reused from an existing source.

### Q9. What tools do data analysts commonly use?

Excel, SQL, Python (Pandas), Power BI, and Tableau are the most common.

### Q10. How do you ensure data quality?

By checking for missing values, duplicates, incorrect types, and inconsistent formatting, and applying validation rules.

**Q11. What is the difference between a metric and a dimension?**

A measure/metric is a numeric value being tracked; a dimension is a categorical field used to slice or group that value.

**Q12. What is descriptive vs predictive analytics?**

Descriptive analytics explains what happened; predictive analytics forecasts what is likely to happen next.

**Q13. How do you handle a tight deadline with incomplete data?**

Prioritize the highest-impact analysis, clearly flag data limitations, and deliver a scoped answer rather than an unreliable full one.

**Q14. What makes a good dashboard?**

It focuses on a few key metrics relevant to its audience, uses consistent design, and is easy to read at a glance.

**Q15. What is the difference between a report and a dashboard?**

A report is typically a static summary of findings; a dashboard is an interactive, often live-updating visual interface.

**Q16. How do you validate your analysis before presenting it?**

By cross-checking totals against a second method (e.g. Excel vs SQL), sanity-checking against known business figures, and reviewing for logical consistency.

**Q17. What is data governance?**

The set of policies and standards that ensure data is accurate, consistent, secure, and used appropriately across an organization.

**Q18. How would you explain a complex finding to a non-technical stakeholder?**

By leading with the business impact in plain language, using a simple visual, and avoiding jargon or unnecessary technical detail.

**Q19. What is the difference between correlation and causation?**

Correlation means two variables move together; causation means one variable directly causes the change in another — correlation alone does not prove causation.

**Q20. Why is domain knowledge important for a data analyst?**

It helps the analyst ask the right questions, interpret results correctly, and avoid conclusions that don't make business sense.

## 18.2 Excel Questions

(20 questions)

**Q1. What is the difference between VLOOKUP and XLOOKUP?**

XLOOKUP can search in any direction and return multiple result types, while VLOOKUP only searches left-to-right and is less flexible.

**Q2. How do you remove duplicate rows in Excel?**

Select the range and use Data > Remove Duplicates, choosing which columns define a duplicate.

**Q3. What is the difference between SUMIF and SUMIFS?**

SUMIF applies one condition; SUMIFS applies two or more conditions simultaneously.

**Q4. What does IFERROR do?**

It returns a custom value instead of an error when a formula fails, such as replacing #DIV/0! with 0.

**Q5. How do you create a Pivot Table?**

Select the data range, go to Insert > PivotTable, then drag fields into Rows, Columns, Values, and Filters.

**Q6. What is conditional formatting used for?**

Automatically highlighting cells based on rules, such as coloring values above a threshold.

**Q7. What is the difference between COUNT and COUNTA?**

COUNT counts only numeric cells; COUNTA counts all non-empty cells regardless of type.

**Q8. How would you find the second-highest value in a range?**

Using =LARGE(range, 2), which returns the nth largest value.

**Q9. What is Data Validation used for?**

Restricting what can be entered into a cell, such as a dropdown list or a numeric range.

**Q10. What does INDEX+MATCH do?**

INDEX returns a value at a given position, and MATCH finds that position — combined, they perform flexible lookups in any direction.

**Q11. How do you split a full name into first and last name?**

Using Text to Columns with a space delimiter, or formulas like LEFT/RIGHT combined with FIND.

**Q12. What is the purpose of an Excel Table (Ctrl+T)?**

It converts a range into a structured, auto-expanding table with built-in filtering and consistent formulas.

**Q13. How do you calculate a running total in Excel?**

Using an absolute-anchored SUM, e.g. =SUM(\$B\$2:B2), copied down the column.

**Q14. What does TRIM do?**

It removes extra spaces from text, keeping single spaces between words.

**Q15. How would you highlight duplicate entries?**

Using Conditional Formatting with a 'Duplicate Values' rule, or COUNTIF-based highlighting.

**Q16. What is the difference between absolute and relative cell references?**

A relative reference (A1) changes when copied to another cell; an absolute reference (\$A\$1) stays fixed.

**Q17. How do you calculate the number of workdays between two dates?**

Using the NETWORKDAYS function, which excludes weekends (and optional holidays).

**Q18. What does the CONCAT/TEXTJOIN function do?**

It combines text from multiple cells into one, with TEXTJOIN also allowing a delimiter and ignoring blanks.

**Q19. How do you protect a worksheet from accidental edits?**

Using Review > Protect Sheet, optionally locking specific cells first via Format Cells.

#### Q20. What is a slicer in Excel?

A visual filter control connected to a Pivot Table or Table, letting users filter data interactively with clickable buttons.

## 18.3 SQL Questions

(30 questions)

#### Q1. What is the difference between WHERE and HAVING?

WHERE filters rows before grouping; HAVING filters groups after aggregation.

#### Q2. What is a primary key vs a foreign key?

A primary key uniquely identifies each row in a table; a foreign key references a primary key in another table to create a relationship.

#### Q3. What is the difference between INNER JOIN and LEFT JOIN?

INNER JOIN returns only matching rows from both tables; LEFT JOIN returns all rows from the left table plus matches from the right.

#### Q4. What is a subquery?

A query nested inside another query, used to compute an intermediate result the outer query depends on.

#### Q5. What is a CTE?

A Common Table Expression, defined with WITH, that creates a named, temporary result set to simplify complex queries.

#### Q6. What is the difference between UNION and UNION ALL?

UNION removes duplicate rows from the combined result; UNION ALL keeps all rows including duplicates.

#### Q7. What does GROUP BY do?

It groups rows sharing the same values in specified columns so aggregate functions can be applied per group.

#### Q8. What is the difference between RANK and DENSE\_RANK?

RANK leaves gaps in the ranking sequence after ties; DENSE\_RANK does not.

#### Q9. What is a window function?

A function that performs a calculation across a set of rows related to the current row, without collapsing them into a single output row.

#### Q10. How do you find duplicate rows in SQL?

Group by the columns that should be unique and use HAVING COUNT(\*) > 1.

#### Q11. What is the difference between DELETE, TRUNCATE, and DROP?

DELETE removes specific rows (can be rolled back); TRUNCATE removes all rows quickly; DROP removes the entire table structure.

#### Q12. What does DISTINCT do?

It removes duplicate rows from the query result, returning only unique combinations of the selected columns.

#### Q13. How do you count unique values in a column?

Using COUNT(DISTINCT column\_name).

**Q14. What is normalization in database design?**

The process of organizing tables to reduce data redundancy and improve data integrity, typically by splitting data into related tables.

**Q15. What is an index and why is it used?**

A database structure that speeds up data retrieval on a column, at the cost of some extra storage and slower writes.

**Q16. What does the LIKE operator do?**

It performs pattern matching in text using wildcards, such as % for any characters and \_ for a single character.

**Q17. How do you handle NULL values in SQL?**

Using IS NULL / IS NOT NULL for comparisons, and COALESCE() to substitute a default value.

**Q18. What is the difference between a view and a table?**

A table stores actual data; a view is a saved, virtual query that displays data dynamically without storing it separately.

**Q19. What does the BETWEEN operator do?**

It filters values within an inclusive range, such as WHERE amount BETWEEN 100 AND 500.

**Q20. How would you calculate a running total in SQL?**

Using a window function: SUM(amount) OVER (ORDER BY order\_date).

**Q21. What is the difference between LAG and LEAD?**

LAG retrieves a value from a previous row; LEAD retrieves a value from a following row, both within a defined ordering.

**Q22. How do you join more than two tables?**

By chaining multiple JOIN clauses, each connecting a new table via a shared key column.

**Q23. What is a self join?**

A join of a table with itself, often used to compare rows within the same table, such as matching employees to their managers.

**Q24. What does the CASE WHEN statement do?**

It applies conditional logic within a query, returning different values based on specified conditions.

**Q25. How do you find the second-highest salary in a table?**

Using OFFSET/FETCH or a subquery with ORDER BY DESC LIMIT 1 OFFSET 1, or DENSE\_RANK() filtered to rank = 2.

**Q26. What is the difference between a clustered and non-clustered index?**

A clustered index determines the physical order of data in a table (only one per table); a non-clustered index is a separate structure pointing to the data.

**Q27. What does the EXISTS operator do?**

It checks whether a subquery returns any rows, often used for efficient existence checks instead of IN.

**Q28. How do you optimize a slow SQL query?**

By adding appropriate indexes, avoiding SELECT \*, filtering early, and reviewing the query execution plan for bottlenecks.

**Q29. What is a foreign key constraint used for?**

It enforces referential integrity, preventing invalid references between related tables.

**Q30. What is the difference between COUNT(\*) and COUNT(column)?**

COUNT(\*) counts all rows regardless of NULLs; COUNT(column) counts only rows where that specific column is not NULL.

## 18.4 Python / Pandas Questions

(20 questions)

**Q1. What is the difference between a list and a tuple?**

Lists are mutable and use square brackets; tuples are immutable and use parentheses.

**Q2. What is a DataFrame?**

A two-dimensional, labeled data structure in Pandas, similar to a spreadsheet or SQL table.

**Q3. How do you handle missing values in Pandas?**

Detect with isnull().sum(), then use dropna() to remove or fillna() to replace missing values.

**Q4. What does groupby() do?**

It splits data into groups based on column values, applies an aggregation, and combines results — the split-apply-combine pattern.

**Q5. What is the difference between loc and iloc?**

loc selects data by label (row/column names); iloc selects data by integer position.

**Q6. How do you merge two DataFrames?**

Using pd.merge(df1, df2, on='key\_column', how='inner'/'left'/'right'/'outer').

**Q7. What is a list comprehension?**

A concise way to build a new list by applying an expression (and optional filter) to each item in an iterable, in one line.

**Q8. What does the apply() function do in Pandas?**

It applies a custom function to each element, row, or column of a DataFrame or Series.

**Q9. How do you convert a column to a datetime type?**

Using pd.to\_datetime(df['column']).

**Q10. What is the difference between merge() and concat()?**

merge() joins DataFrames on a key column like a SQL join; concat() simply stacks DataFrames vertically or horizontally.

**Q11. How do you filter rows based on multiple conditions?**

Using boolean indexing with & (and) or | (or), e.g. df[(df.a > 1) & (df.b == 'X')].

**Q12. What does df.describe() return?**

Summary statistics (count, mean, std, min, quartiles, max) for numeric columns.

**Q13. How do you remove duplicate rows in Pandas?**

Using df.drop\_duplicates().

**Q14. What is the difference between a Series and a DataFrame?**

A Series is a single labeled column; a DataFrame is a table made of multiple Series sharing an index.

**Q15. How would you find the top 5 rows by a column's value?**

Using df.nlargest(5, 'column') or df.sort\_values('column', ascending=False).head(5).

**Q16. What is exception handling and why use it?**

Using try/except to gracefully handle unexpected errors, such as bad data or missing files, without crashing the script.

**Q17. How do you rename columns in a DataFrame?**

Using df.rename(columns={'old':'new'}).

**Q18. What is broadcasting in NumPy?**

Automatic expansion of a smaller array's shape so it can be combined element-wise with a larger array in arithmetic operations.

**Q19. How do you read a CSV file into Pandas?**

Using pd.read\_csv('filename.csv').

**Q20. What does the pivot\_table() function do in Pandas?**

It reshapes and summarizes data by aggregating values across rows and columns, similar to an Excel Pivot Table.

## 18.5 Statistics Questions

(20 questions)

**Q1. What is the difference between mean, median, and mode?**

Mean is the arithmetic average, median is the middle value when sorted, and mode is the most frequently occurring value.

**Q2. What is standard deviation?**

A measure of how spread out data is around the mean; the square root of variance.

**Q3. What is the difference between population and sample?**

A population includes every member of the group being studied; a sample is a subset drawn from it.

**Q4. What is a p-value?**

The probability of observing results as extreme as the data, assuming the null hypothesis is true; a small p-value suggests rejecting the null hypothesis.

**Q5. What is the difference between correlation and causation?**

Correlation shows two variables move together; causation means one directly causes the other — correlation alone does not prove causation.

**Q6. What is a normal distribution?**

A symmetric, bell-shaped distribution where most values cluster around the mean.

**Q7. What is the Interquartile Range (IQR) used for?**

Measuring the spread of the middle 50% of data and detecting outliers using the  $1.5 \times \text{IQR}$  rule.

**Q8. What is the difference between Type I and Type II errors?**

A Type I error rejects a true null hypothesis (false positive); a Type II error fails to reject a false null hypothesis (false negative).

**Q9. What is a confidence interval?**

A range that is likely to contain the true population value, at a stated confidence level such as 95%.

**Q10. What is skewness?**

A measure of the asymmetry of a distribution — right skew has a long right tail, left skew has a long left tail.

**Q11. What is the null hypothesis?**

The default assumption of no effect or no difference, which a statistical test attempts to find evidence against.

**Q12. What is the difference between variance and standard deviation?**

Variance is the average squared deviation from the mean; standard deviation is its square root, expressed in the original units.

**Q13. What is sampling bias?**

A systematic error where the sample does not represent the population, leading to misleading conclusions.

**Q14. What does statistical significance mean?**

That an observed result is unlikely to have occurred by random chance alone, typically judged against a p-value threshold.

**Q15. What is the difference between discrete and continuous data?**

Discrete data takes specific countable values; continuous data can take any value within a range.

**Q16. What is covariance?**

A measure of the direction of the relationship between two variables, though unlike correlation it is not standardized.

**Q17. What are outliers and how are they identified?**

Extreme values far from the rest of the data, often identified using the IQR rule or z-scores beyond  $\pm 3$ .

**Q18. What is stratified sampling?**

A sampling method where the population is divided into subgroups and samples are drawn proportionally from each.

**Q19. What is the Central Limit Theorem (in simple terms)?**

It states that the distribution of sample means approaches a normal distribution as sample size increases, regardless of the population's original distribution.

#### **Q20. Why is a large sample size generally preferred?**

It reduces sampling error and produces more reliable, generalizable estimates of the population.

## **18.6 Power BI Questions**

(20 questions)

#### **Q1. What is the difference between Power Query and DAX?**

Power Query imports and transforms data before modeling; DAX is used to write calculations on data already in the model.

#### **Q2. What is a measure in Power BI?**

A calculation evaluated dynamically at query time based on the current filter context, typically used for aggregations.

#### **Q3. What is a calculated column?**

A column computed once at data refresh and stored in the table, typically used for row-level classification.

#### **Q4. What is CALCULATE used for?**

It changes the filter context under which an expression is evaluated, enabling conditional aggregations.

#### **Q5. What is a star schema?**

A model with one central fact table connected to multiple dimension tables, improving performance and clarity.

#### **Q6. What is the difference between a fact table and a dimension table?**

A fact table holds transactional, numeric data; a dimension table holds descriptive attributes used to filter and group facts.

#### **Q7. What is a slicer?**

An on-report visual filter control that lets users interactively filter the data shown in a dashboard.

#### **Q8. How do you create a relationship between two tables?**

By connecting a common key column between tables in the Model view, defining cardinality and filter direction.

#### **Q9. What is time intelligence in DAX?**

A set of functions (like TOTALYTD, SAMEPERIODLASTYEAR) for analyzing data across custom time periods.

#### **Q10. What is the difference between SUM and SUMX?**

SUM aggregates a single column directly; SUMX evaluates an expression row-by-row before summing, allowing more complex calculations.

#### **Q11. What is Row-Level Security (RLS) used for?**

Restricting which rows of data specific users can see within the same report.

#### **Q12. What does FILTER do in DAX?**

It returns a filtered table based on a specified condition, often used inside CALCULATE.

#### **Q13. How do you publish a Power BI report?**

Using the 'Publish' button in Power BI Desktop to send the report to a workspace in the Power BI Service.

**Q14. What is the difference between import and DirectQuery mode?**

Import mode loads data into Power BI's in-memory model for fast performance; DirectQuery queries the source live without importing data.

**Q15. What is a hierarchy in Power BI?**

A structured grouping of related fields (e.g. Year > Quarter > Month) that enables drill-down in visuals.

**Q16. What is the purpose of the 'Fields' pane vs the 'Filters' pane?**

The Fields pane lists available data columns/measures; the Filters pane applies scoped filtering at the visual, page, or report level.

**Q17. How would you improve a slow-loading Power BI report?**

By reducing model size, optimizing DAX measures, limiting visuals per page, and using aggregated tables where possible.

**Q18. What is a bookmark in Power BI?**

A saved snapshot of a report's current filter and visual state, used to build interactive navigation or presentations.

**Q19. What does DISTINCTCOUNT do?**

It counts the number of unique values in a column, useful for metrics like unique customers.

**Q20. What is the difference between a report and a dashboard in the Power BI Service?**

A report contains one or more interactive pages built in Desktop; a dashboard is a single-page collection of pinned visuals, often from multiple reports.

**Key Takeaways**

- Interviewers value concise, structured answers over long explanations.
- Practice explaining SQL joins and DAX/Excel formula differences out loud.
- Be ready to walk through one real project end-to-end in detail.
- Statistics fundamentals (mean, std dev, p-value) come up in nearly every interview.

# Data Analyst Portfolio and Career Roadmap

## Learning Objectives

- Understand the skills roadmap from beginner to job-ready.
- Learn how to build a portfolio that demonstrates real skill.
- Prepare a resume, LinkedIn profile, and interview presentation.
- Follow a structured 90-day learning plan.

## 19.1 How to Become a Data Analyst

Becoming job-ready as a data analyst is less about credentials and more about demonstrable skill: the ability to take a messy dataset and produce a clear, correct, well-communicated answer to a business question. This is built through structured learning followed by real project practice.

## 19.2 Skills Roadmap: Beginner to Advanced

| Stage        | Focus                                                             |
|--------------|-------------------------------------------------------------------|
| Beginner     | Excel fundamentals, basic SQL, core statistics                    |
| Intermediate | Advanced SQL, Python/Pandas, EDA, Power BI or Tableau             |
| Advanced     | End-to-end projects, storytelling, portfolio, interview readiness |

## 19.3 Building Projects

- Choose projects with a realistic business question, not just a random dataset.
- Cover the full workflow: cleaning, analysis, visualization, and a written conclusion.
- Include at least one SQL-heavy project and one Python/Pandas-heavy project.
- Include at least one Power BI or Tableau dashboard project.

## 19.4 Building a GitHub Portfolio

- 1 Create a GitHub account and one repository per project.
- 2 Include the dataset (or a link to it), code/notebooks, and a clear README.
- 3 In the README, describe: the business problem, approach, key findings, and recommendations — not just the code.
- 4 Add screenshots of dashboards or key charts directly in the README.
- 5 Keep code clean and commented so a recruiter can follow your thinking.

## 19.5 Creating a Resume

- Lead with impact: 'Analyzed sales data to identify a \$40K profit leak' beats 'Used Excel and SQL'.
- List tools (Excel, SQL, Python, Power BI/Tableau) in a dedicated skills section.
- Include 2–4 projects with a one-line result each.
- Keep it to one page for early-career roles.

## 19.6 LinkedIn Profile

- Use a headline that states your target role, e.g. 'Aspiring Data Analyst | SQL, Python, Power BI'.
- Write a short 'About' section summarizing your skills and project focus.
- Post or share project write-ups to build visibility.

## 19.7 Interview Preparation

Revisit Chapter 18 regularly, but also rehearse presenting one complete project out loud in under five minutes: the problem, your approach, the key finding, and the recommendation.

## 19.8 How to Present Projects

- 1 State the business problem in one sentence.
- 2 Briefly describe the dataset and any major cleaning challenges.
- 3 Show one or two key charts, not every chart you made.
- 4 State the main insight clearly.
- 5 End with a specific, actionable recommendation.

## 19.9 Common Mistakes Beginners Make

- Jumping into charts before defining a clear business question.
- Skipping data cleaning and presenting results built on dirty data.
- Building portfolios with generic, overused datasets and no original angle.
- Focusing only on tools, not on communicating insights clearly.
- Not quantifying impact ('improved reporting' vs 'cut reporting time by 60%').

## 19.10 90-Day Data Analyst Learning Roadmap

| Days  | Focus                                                       |
|-------|-------------------------------------------------------------|
| 1–15  | Excel fundamentals + core statistics                        |
| 16–30 | SQL — from SELECT basics through joins and window functions |
| 31–45 | Python + NumPy + Pandas fundamentals and data cleaning      |
| 46–55 | Exploratory Data Analysis and data visualization principles |

| Days  | Focus                                                             |
|-------|-------------------------------------------------------------------|
| 56–70 | Power BI or Tableau — modeling, DAX/calculated fields, dashboards |
| 71–80 | Build 2 full portfolio projects end-to-end                        |
| 81–90 | Resume, LinkedIn, GitHub polish, and mock interview practice      |

### Motivation

Most working data analysts did not start with a data science degree — they built consistent daily practice and a small portfolio of real, well-explained projects. Focus on depth over breadth: two excellent projects beat ten shallow ones.

### Key Takeaways

- A strong portfolio demonstrates the full workflow, not just isolated skills.
- GitHub READMEs should explain the business story, not only show code.
- Resumes should quantify impact, not just list tools used.
- A 90-day structured plan is enough to go from fundamentals to job-ready.

## Practice Questions

- 1 Draft a one-paragraph GitHub README summary for a project you have completed or plan to build.
- 2 Rewrite the resume line 'Used Excel and SQL for reporting' to lead with impact.
- 3 Outline a five-minute project presentation using the structure in section 19.8.

# Quick Revision

## Learning Objectives

- Refresh every essential definition, formula, and command before an exam or interview.

## Important Definitions

| Term            | Definition                                                      |
|-----------------|-----------------------------------------------------------------|
| Data Analytics  | Examining data to draw conclusions and support decisions        |
| KPI             | A metric tied directly to a strategic business goal             |
| Structured Data | Data organized in a fixed schema of rows and columns            |
| Outlier         | A value far removed from the rest of the dataset                |
| Normalization   | Rescaling data into a fixed range, typically 0 to 1             |
| p-value         | Probability of the observed data if the null hypothesis is true |
| Star Schema     | A fact table connected to multiple dimension tables             |

## Important Formulas

### Formula

Mean = Sum of values ÷ Number of values

### Formula

Standard Deviation ( $\sigma$ ) =  $\sqrt{(\sum(x - \text{mean})^2 \div n)}$

### Formula

IQR = Q3 – Q1 (Outlier if beyond Q1 – 1.5×IQR or Q3 + 1.5×IQR)

### Formula

Conversion Rate = (Conversions ÷ Total Visitors) × 100

### Formula

CAC = Total Marketing & Sales Cost ÷ New Customers Acquired

### Formula

CLV = Average Order Value × Purchase Frequency × Customer Lifespan

#### Formula

$\text{Churn Rate} = (\text{Customers Lost} \div \text{Customers at Start}) \times 100$

#### Formula

$\text{Profit Margin} = (\text{Profit} \div \text{Sales}) \times 100$

## Important Excel Functions

- SUMIFS / COUNTIFS / AVERAGEIF — conditional aggregation
- XLOOKUP / INDEX+MATCH — flexible lookups
- IF / IFERROR / AND / OR — logical formulas
- Pivot Tables — instant summarization without manual formulas

## Important SQL Commands

- SELECT ... WHERE ... GROUP BY ... HAVING ... ORDER BY
- JOIN types: INNER, LEFT, RIGHT, FULL, SELF
- Window functions: ROW\_NUMBER, RANK, DENSE\_RANK, LAG, LEAD
- WITH ... AS (CTE) for readable, multi-step queries

## Important Python / Pandas Functions

- df.head(), df.info(), df.describe() — quick inspection
- df.dropna(), df.fillna(), df.drop\_duplicates() — cleaning
- df.groupby().agg() — split-apply-combine summarization
- pd.merge() — SQL-style joins between DataFrames

## Statistics Formulas Recap

- Variance = average of squared deviations from the mean
- Correlation ranges from -1 to +1; measures linear relationship strength
- Confidence interval: a range likely to contain the true population value

## Power BI / DAX Concepts Recap

- Measures calculate dynamically; calculated columns are fixed at refresh
- CALCULATE changes the filter context of an expression
- Star schema: one fact table, multiple dimension tables

## Visualization Rules Recap

- Match chart type to the question: comparison, trend, distribution, or relationship
- Never truncate a bar chart's y-axis without a clear label
- Keep dashboards focused on a handful of key metrics

## Interview Preparation Checklist

- Can explain the Data Analytics Lifecycle end-to-end
- Comfortable writing SQL joins, GROUP BY, and window functions from memory
- Can explain mean/median/mode, standard deviation, and p-value simply
- Can walk through one full project in under five minutes
- Has a polished resume, LinkedIn profile, and GitHub portfolio ready

DATA-ANALYTICS-HUB

# Final Project Checklist

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## Learning Objectives

- Confirm readiness across every core skill covered in this course.

## Can You Do the Following?

- ☐ Clean a raw, messy dataset (missing values, duplicates, formatting)
- ☐ Analyze data in Excel using formulas and Pivot Tables
- ☐ Write SQL queries including joins, aggregations, and window functions
- ☐ Write basic-to-intermediate Python for data tasks
- ☐ Use Pandas to load, clean, and summarize a dataset
- ☐ Perform Exploratory Data Analysis and interpret the results
- ☐ Create clear, honest data visualizations
- ☐ Build an interactive Power BI or Tableau dashboard
- ☐ Explain business insights clearly to a non-technical audience
- ☐ Complete an end-to-end analytics project from question to recommendation

If you can confidently check every box above, you are ready to apply for Data Analyst roles and to discuss your projects in interviews with confidence. Revisit any chapter where you hesitated before moving on.